with the following conditions on function value and first and second derivatives:

$$\begin{aligned} f_i(x_i) &= y_i \qquad i = 1, \ldots, n-1 \\ f_{i-1}(x_i) &= y_i \qquad i = 2, \ldots, n \end{aligned} \tag{20.9.2}$$

$$f'_{i-1}(x_i) = f'_i(x_i) \qquad i = 2, \ldots, n-1 \tag{20.9.3}$$

$$f''_{i-1}(x_i) = f''_i(x_i) \qquad i = 2, \ldots, n-1 \tag{20.9.4}$$

The individual cubic polynomial $f_i(x)$ can be expressed using the interval values y_i, y_{i+1} and either y'_i, y'_{i+1}, or y''_i, y''_{i+1} to represent the cubic coefficients. Both forms yield the same $y(x)$, of course, but since using the mesh-point second-derivative values as unknowns has some minor advantages in most practical computation, the very similar derivation of the slope form is omitted here.

Assuming $y'''(x)$ = constant in each interval means that $y''(x)$ is linear, we get

$$f''_i(x) = y''_i\left(\frac{x_{i+1} - x}{h_i}\right) + y''_{i+1}\left(\frac{x - x_i}{h_i}\right) \tag{20.9.5}$$

Integrating twice more and selecting the constant of integration such that Eqs. (20.9.2) are satisfied yields

$$\begin{aligned} f_i(x) = {} & y_i\left(\frac{x_{i+1} - x}{h_i}\right) + y_{i+1}\left(\frac{x - x_i}{h_i}\right) \\ & - \frac{h_i^2}{6} y''_i\left[\frac{x_{i+1} - x}{h_i} - \left(\frac{x_{i+1} - x}{h_i}\right)^3\right] \\ & - \frac{h_i^2}{6} y''_{i+1}\left[\frac{x - x_i}{h_i} - \left(\frac{x - x_i}{h_i}\right)^3\right] \end{aligned} \tag{20.9.6}$$

which identically satisfies the continuity condition [Eq. (20.9.4)] on the second derivative. This form does not, in general, satisfy the continuity conditions on slope [Eq. (20.9.3)]. Expanding the LHS and RHS of Eq. (20.9.3) by differentiating Eq. (20.9.6) and evaluating yields, respectively,

$$f'_i(x_i) = \frac{y_{i+1} - y_i}{h_i} - \frac{h_i}{6}(2y''_i + y''_{i+1}) \tag{20.9.7}$$

$$f'_{i-1}(x_i) = \frac{y_i - y_{i-1}}{h_{i-1}} + \frac{h_{i-1}}{6}(y''_{i-1} + 2y''_i) \tag{20.9.8}$$

which, when equated, produce the condition

$$h_{i-1}y''_{i-1} + 2(h_{i-1} + h_i)y''_i + h_i y''_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right) \qquad i = 2, \ldots, n-1 \tag{20.9.9}$$

which must be satisfied at $n-2$ points by the n unknown quantities y_i''. Two more conditions are required on the y_i'', and these are obtained by specifying one end condition at each end.

Specifying the end second derivatives y_1'' and y_n'' leaves $n-2$ unknowns, which can be obtained by solving the $n-2$ equations of Eq. (20.9.9). The coefficient matrix is tridiagonal, and the diagonal term is dominant, which assures us that the matrix is never singular. An example of Eq. (20.9.9) in matrix form is shown for six equally spaced points, where $d_i = (y_{i+1} - 2y_i + y_{i-1})/(2h^2)$:

$$\begin{pmatrix} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} y_2'' \\ y_3'' \\ y_4'' \\ y_5'' \end{pmatrix} = \begin{pmatrix} 12d_2 - \dfrac{y_1''}{h} \\ 12d_3 \\ 12d_4 \\ 12d_5 - \dfrac{y_6''}{h} \end{pmatrix} \qquad (20.9.10)$$

Instead of specifying y_1'', we could specify the end slope y_1'—or a linear combination of the two. If y_1' is specified, then y_1'' is an unknown, and we must find an equation involving both. Evaluating Eq. (20.9.7) for $i = 1$ yields the required equation,

$$2h_1 y_1'' + h_1 y_2'' = 6\left(\frac{y_2 - y_1}{h_1} - y_1'\right) \qquad (20.9.11)$$

which, when included in the above example, alters Eq. (20.9.10) slightly to the form

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 0 \\ 0 & 0 & 1 & 4 & 1 \\ 0 & 0 & 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} y_1'' \\ y_2'' \\ y_3'' \\ y_4'' \\ y_5'' \end{pmatrix} = \begin{pmatrix} 12d_1 \\ 12d_2 \\ 12d_3 \\ 12d_4 \\ 12d_5 - \dfrac{y_6''}{h} \end{pmatrix}$$

where $d_1 \equiv (y_2 - y_1 - y_1'h)/(2h^2)$. If y_n' is specified instead of y_n'', an equation similar to Eq. (20.9.11) appears at the bottom of the matrix.

If the appropriate end conditions are not actually known, the simplest choice is $y_1'' = 0$. Another, and smoother, choice is $y_1'' = 0.5y_2''$.

Although the spline is a "global" rather than a "local" curve, i.e., altering a single y_i or end condition affects the spline throughout $[x_1, x_n]$, the dominant diagonal term causes such effects to become small rapidly as the distance from the altered point increases, virtually eliminating roundoff problems. In addition, since the solution of the tridiagonal equations involves βn arithmetic operations, there are no computational difficulties or disadvantages in fitting hundreds of points at a time.

The storage of the spline coefficients y_i, y_i'' $(i = 1, \ldots, n)$ involves $2n$ locations compared with only n locations for the coefficients of an n-point interpolation polynomial.

20.10 COMPARISON WITH POLYNOMIAL INTERPOLATION

The results of the widely used polynomial interpolation are frequently perfectly acceptable, as illustrated in Fig. 20.10.1. Generally speaking, the spline does not have any particular advantages when used for (1) the approximation of a well-behaved, easily calculated mathematical function (unless the derivatives are also important) or (2) the curve-fitting of experimental data which are "dense." The term *dense* will be loosely defined here as meaning that the number of points, in any subregion, is more than an order of magnitude larger than the number of inflection points expected in the fitted curve for that subregion, and that there are no "abrupt" changes in the expected second derivative.

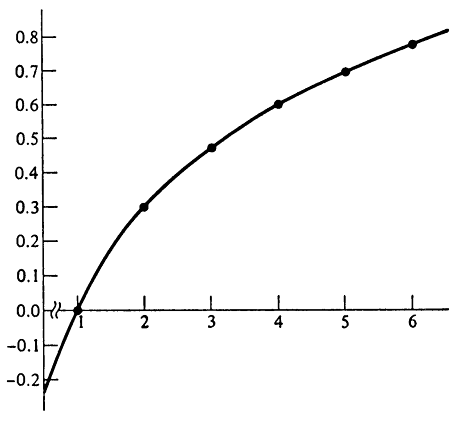

FIGURE 20.10.1
An example of smooth polynomial interpolation.

The spline function's smoothness properties give it significant advantages over polynomials when dealing with "sparse" data; i.e., the number of points is less than an order of magnitude larger than the number of expected inflection points, or there is a large change in the expected second derivative over a small number of points.

One of the polynomial characteristics which causes great difficulty is the manner in which an oscillation caused by one bad point or "difficult" region is reflected, usually undiminished, throughout $[x_1, x_n]$. Figure 20.10.2 shows such an example.

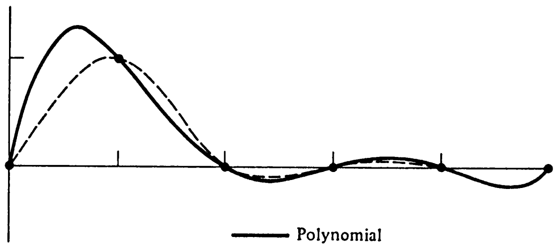

FIGURE 20.10.2
Comparison of oscillation characteristics.

All but one of the six points are in a straight line, but the solid line corresponding to the interpolating polynomial continues oscillating with an amplitude which would not diminish, even if there were 50 more data points in a line to the right. The amplitudes of the oscillation for the interpolating cubic spline function (with $y_1'' = y_n'' = 0$), indicated by the dashed line, are reduced by about 1/3 in each successive interval.

Frequently in engineering problems the first or second derivative of the fitted curve is of more interest than the function value. Figure 20.10.3 is an example of sparse data for which the second derivative was of interest. Probably largely due to the point at x_5, the interpolating polynomial (Fig. 20.10.3) is unacceptable, giving large excursions of the second derivative and, as we shall discuss later, an unnecessary number of them. It should be emphasized that the fifth point was not a piece of bad data but, in fact, indicated a legitimate localized oscillation of considerable interest. Of course it is easy to dismiss this by insisting that more points should have been measured in that region. However, one cannot dismiss the feeling that something smoother can be fitted to these points.

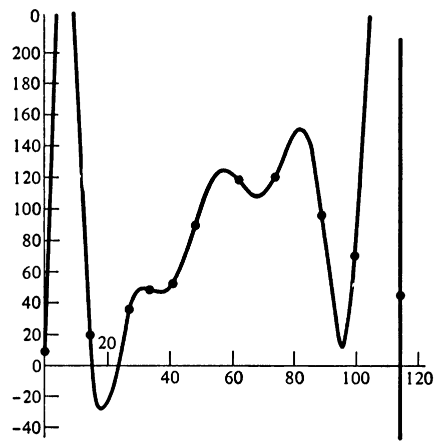

FIGURE 20.10.3
Polynomial interpolation of data.

The interpolation spline for these data (with $y_1'' = y_n'' = 0$), shown in Fig. 20.10.4, demonstrates that the high-degree continuity associated with polynomials should not be confused with the smoothness required in the typical physical problem. We are, in fact, sacrificing higher-derivative continuity to obtain second-derivative smoothness. Where third derivative smoothness is important, we note that the quintic spline function (Ref. 2, pp. 143–148) minimizes $\int (y''')^2\,dx$, and similarly for higher odd-degree splines. The resulting degradation of second-derivative smoothness is unacceptable in the typical physical problem, but the higher odd-integer spline functions are of interest in function approximation since all nonzero derivatives of a spline converge to the corresponding function derivative (Ref. 2, pp. 135–143).

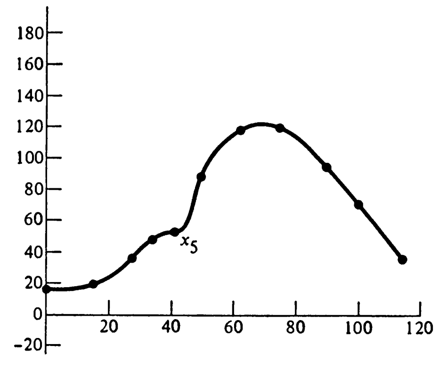

FIGURE 20.10.4
Spline interpolation of data in Fig. 20.10.3.

The possible failure of polynomial interpolation can be anticipated by looking at a divided difference table (Sec. 18.2). Differences of the jth order can be associated with the jth derivative of an interpolating polynomial through $j + 1$ adjacent points. Scanning a column in the difference table gives an indication of how stable the corresponding derivative should be. If the higher-order differences do not become smooth, polynomial interpolation will likely be unacceptable. The difference table for the points in Fig. 20.10.3 is shown in Table 20.10. The strong plus-minus oscillation in the fourth-difference column is a sure sign of trouble.

While the term *smoothness* is mathematically vague, there is one necessary condition which can be formally defined and is useful here in comparing the spline and polynomial.

The points in Fig. 20.10.1 have negative second differences. An obvious requirement for a smooth interpolating function is to have inflection points only where the second differences change sign.[1] In Table 20.10 the second differences have four changes of sign, while the interpolating polynomial (Fig. 20.10.3) has eight inflection points—four *extraneous* inflection points. The spline (Fig. 20.10.4) has four inflection points, satisfying this necessary smoothness requirement. While in practice the cubic spline almost always satisfies this requirement, there exist data for which it will fail. The more general spline-in-tension function always succeeds and is useful in these exceptional cases.

[1] D. G. Schweikert, An Interpolation Curve Using a Spline in Tension, *J. Math. Phys.*, 45, pp. 312–317, 1966.

Table 20.10 DIVIDED DIFFERENCE TABLE FOR DATA IN FIG. 20.10.3

x	y	1st	2nd $\times 10^1$	3rd $\times 10^2$	4th $\times 10^3$	5th $\times 10^5$	6th $\times 10^6$	7th $\times 10^7$	8th $\times 10^9$	9th $\times 10^{11}$	10th $\times 10^{12}$
0	16										
		0.2									
14	19		0.4								
		1.3		−0.0							
27	36		0.4		−0.12						
		2.0		−0.5		1.7					
33	48		−1.0		0.72		−0.86				
		0.6		2.0		−3.6		0.27			
41	53		3.1		−1.01		1.12		−0.56		
		5.3		−1.6		3.1		−0.23		0.98	
48	90		−1.5		0.45		−0.64		0.40		−0.13
		2.1		0.2		−0.9		0.11		−0.55	
62	119		−0.8		−0.04		0.14		−0.15		
		0.1		0.0		0.1		−0.02			
74	120		−0.6		0.00		−0.00				
		−1.6		0.1		0.0					
89	96		−0.4		0.00						
		−2.5		0.1							
99	71		0.1								
		−2.3									
114	36										

21

INDEFINITE INTEGRALS—FEEDBACK

21.1 INTRODUCTION

There is a significant difference between computing a definite integral

$$\int_a^b f(x)\,dx$$

and computing an indefinite integral, which may be written in the form

$$\int_a^x f(x)\,dx$$

The result of the first computation is a single number, while the result of the second is a table of numbers.

The main purpose of this chapter is to introduce, after a brief survey, two new concepts. The first concept is that of *instability*, which can occur when function values that have already been computed are used to compute the next value. This repeated use of the function values may be compared to "feedback" and produces instability in the same manner. Feedback is the use of part of the output at a particular time as input at a slightly later time. Sometimes this time delay can be neglected, and sometimes it cannot. Perhaps the simplest physical example is the feedback amplifier which is widely used in control circuits. (See Fig. 21.1.1.) The output voltage is the sum of the two inputs on the left times the gain of the amplifier, which is taken as -10^9, and one-tenth of the output is fed into the input.

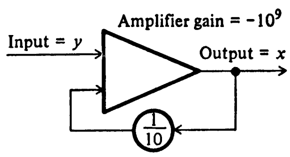

FIGURE 21.1.1

We have

$$(y + \tfrac{1}{10}x)(-10^9) = x$$

or

$$x = -\frac{10^{10}y}{10^9 + 10} = -\frac{10y}{1 + 10^{-8}} \approx -10y$$

Thus the output is approximately -10 times the input, quite independent of small changes in the characteristics of the amplifier; that is, it is not sensitive to the exact value of the gain.

Under some circumstances, if there is enough delay in returning the signal from the output to the input, the system oscillates—*sings* is the word often used. Such situations arise frequently in public auditoriums which have a microphone (y) and an amplifier system output (x). The output from the loudspeaker is delayed by the time that it takes the sound to get from the speaker to the microphone, and the result is heard by the audience unless care is taken to control it. The same unstable situation sometimes occurs in adjusting a shower to the desired temperature.

The theory of feedback systems is well developed and cannot be treated in detail here. We stress the point that when we use an output result as an input to the same process at a later stage, we run the risk of producing an unwanted oscillation, whose period and behavior depend on the particular process.

The second new concept is an extension of the technique that we have been using to find formulas. In the past we have used all the parameters that we had in order to make the formula exact for as many powers of x as we could; or, in other words, we have concentrated on making the truncation error of as high an order as possible. In order to cope with instability and other effects, such as roundoff propagation, it is necessary to leave some of the coefficients of the general formula undetermined at the time we are finding the error term, and later select values for them which will produce suitable formulas.

The problem of calculating an indefinite integral

$$y(x) = \int_a^x f(x)\, dx$$

may be recast in the form of a simple differential equation

$$y'(x) = f(x) \qquad y(a) = 0$$

In this form the integrand values are labeled y', while the answer is labeled y. We shall adopt the differential equation approach because it blends in with later notation when the more general differential equation

$$y'(x) = f(x, y) \qquad y(a) = b$$

is examined. (See Chap. 23.) This approach somewhat limits the class of formulas that we find.

We shall also make a slight change in our notation. Up to this time we have been using $y(x)$ and y_x to mean the same thing. We shall now use the subscript notation to count; that is, we shall number the points at which we compute the answer $0, 1, 2, \ldots$. But for reasons of clarity it is desirable to keep the spacing of the problem at h and to let the origin fall where it may. Thus $y(a + nh)$ and y_n are the same quantity. No confusion should arise once the convention is understood, and the notation becomes a great deal simpler.

21.2 SOME SIMPLE FORMULAS FOR INDEFINITE INTEGRALS

A simple formula for approximating one step of an indefinite integral is

$$y_{n+1} = y_n + hy'[a + (n + \tfrac{1}{2})h] = y_n + hy'_{n+1/2}$$

where

$$y' = f(x)$$

or

$$y = \int_a^x y'\, dx$$

This is clearly the midpoint formula, and it has one-half the error of the usual trapezoid rule

$$y_{n+1} = y_n + \frac{h}{2}(y'_{n+1} + y'_n)$$

The midpoint formula also has better roundoff properties.

A more accurate formula, which may be found by using the universal matrices of Sec. 15.6, is

$$y_{n+1} = y_n + \frac{h}{24}(-y'_{n-1} + 13y'_n + 13y'_{n+1} - y'_{n+2})$$

where the error term is

$$\tfrac{11}{720}h^5 y^{(5)}(\theta) \approx 0.0153 h^5 y^{(5)}(\theta)$$

If this formula is used[1] to compute an integral, then two extra points which lie outside the range of integration have to be computed (one at each end). If the values of the integrand cannot be found at such points because of singularities in the formula, then the assumption that the integrand can be approximated accurately by a polynomial is probably false. An alternative formula can be found for the ends of the interval, namely,

$$y_1 = y_0 + \frac{h}{24}(9y'_0 + 19y'_1 - 5y'_2 + y'_3)$$

Simpson's formula is widely used for computing indefinite integrals. In our present notation this takes the form

$$y_{n+1} = y_{n-1} + \frac{h}{3}(y'_{n-1} + 4y'_n + y'_{n+1})$$

with an error term

$$-\frac{h^5 y^{(5)}(\theta)}{90} \approx -0.0111 h^5 y^{(5)}(\theta)$$

which is slightly less than the above formula.

There is one often overlooked trouble that occurs when Simpson's formula is used in the conventional manner, and although the trouble is not serious, it can be annoying. Simpson's formula carries the value of the integral two steps forward, and the special Simpson's half-formula (Sec. 15.6) is used to start the chain for the odd-numbered sample points. The result of this jumping two steps ahead is that the accumulated errors at the odd- and even-numbered points are rather independent of each other, especially as the weights attached to the computed integrand values are different. All this tends to produce an oscillation. While this oscillation is due to errors being committed, and hence gives some

[1] We of course actually compute $(24/h)y_n$ and multiply by $h/24$ when we print out values of y_n. In this way, while we do the same amount of computing, the roundoff effects do not build up so rapidly, since the final multiplication by $h/24$ does not propagate further.

measure of the accuracy of the results, it can be very annoying at times. The oscillation can be avoided if only the even-numbered-point chain is computed and Simpson's half-formula is used to produce each odd-numbered point.

PROBLEMS

21.2.1 Derive the four-point integration formulas given in this section with their error terms.

21.3 A GENERAL APPROACH

Many special formulas could be investigated one at a time, but we shall turn to a general approach and examine a whole class at once. In estimating the next value y_{n+1} of the integral, we can use old values of the integral $y_n, y_{n-1}, \ldots$, as well as both new values of the integrand $y'_{n+1}, \ldots$ and old values $y'_n, y'_{n-1}, \ldots$. We shall arbitrarily limit the present study to the form

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2}) + E_5 \frac{h^5 y^{(5)}(0)}{5!}$$

which uses three old values of the integral together with one new and three old values of the integrand. The question immediately arises, Why not try the more reasonable approach of using integrand values symmetrically placed about the value being computed? Such an approach would probably give more accurate formulas, but we shall not do so for reasons that will become apparent when we investigate the general differential equation.

We have written the equations so that the defining equations will be homogeneous in h; *we can therefore act as if* $h = 1$, most of the time, and *remember* to supply the h factors at the proper moment. This involves a slight change in the meaning of E_5, but no serious confusion will occur in practice.

With seven parameters in the general form, we shall impose only five conditions—namely, that it be exact for $y = 1, x, \ldots, x^4$, or, what is the same thing, exact for $y = 1, (x - n), \ldots, (x - n)^4$—and we shall save two parameters for other purposes. Earlier we made the formula exact when the *integrand* was $1, x, x^2, \ldots$; now we make the *answer* exact for $1, x, x^2, \ldots$. Since the integrals of powers of x are powers one higher in x, we need account only for why we have added the condition of exactness for $y = 1$. The consequence of not doing so is that $y' = 0$ will not integrate into $y =$ constant. Using these five conditions and taking a_1 and a_2 as parameters, we get from the defining equations

$$\begin{aligned} a_0 &= 1 - a_1 - a_2 & b_0 &= \tfrac{1}{24}(19 + 13a_1 + 8a_2) \\ a_1 &= a_1 & b_1 &= \tfrac{1}{24}(-5 + 13a_1 + 32a_2) \\ a_2 &= a_2 & b_2 &= \tfrac{1}{24}(1 - a_1 + 8a_2) \\ b_{-1} &= \tfrac{1}{24}(9 - a_1) & E_5 &= \tfrac{1}{6}(-19 + 11a_1 - 8a_2) \end{aligned}$$

Note that the equations are linear in a_1 and a_2. Each point in the a_1, a_2 plane corresponds to a formula—a common mathematical device for studying families of formulas.

PROBLEMS

21.3.1 Derive the above formulas for the coefficients.

21.4 TRUNCATION ERROR

We have written the general form of our formula as if the influence function $G(s)$ (see Chap. 16) had a constant sign (Sec. 16.5). It is necessary to examine this assumption and find which values of a_1, a_2 make this true.

Using Chap. 16 as our guide, we first notice that, except for special values of a_1, a_2 which make $E_5 = 0$, we have $m = 5$. The influence function of Sec. 16.4 has an $A = -2h$, so that the Taylor expansion is (Sec. 16.2)

$$\begin{aligned} y(x) &= y(-2h) + (x + 2h)y'(-2h) + \frac{(x+2h)^2}{2!}y''(-2h) \\ &\quad + \frac{(x+2h)^3}{3!}y'''(-2h) + \frac{(x+2h)^4}{4!}y^{(4)}(-2h) \\ &\quad + \frac{1}{4!}\int_{-2h}^{x} y^{(5)}(s)(x - s)^4\,ds \end{aligned}$$

If this is substituted in the general form of Sec. 21.3, we pick $B = h$ and get

$$R(y) = \int_{-2h}^{h} y^{(5)}(s)G(s)\,ds$$

where $G(s)$ is the influence function. (Note that no integration is needed in constructing it.)

$$\begin{aligned} G(s) = \frac{1}{4!}\{&(h - s)_+^4 - a_0(-s)_+^4 - a_1(-h - s)_+^4 - a_2(-2h - s)_+^4 \\ &- 4h[b_{-1}(h - s)_+^3 + b_0(-s)_+^3 + b_1(-h - s)_+^3 + b_2(-2h - s)_+^3]\} \end{aligned}$$

Now, if $G(s)$ is of constant sign (Sec. 16.5), then we have

$$R(x^5) = 5! \int_{-2h}^{h} G(s)\,ds = E_5 h^5$$

where E_5 is the result of substituting $y(x) = x^5$ for unit spacing.

The problem is to find those values of a_1, a_2 such that $G(s)$ has a constant sign. Thus we need to find the zeros of $G(s)$ for each pair (a_1, a_2). Since this is difficult to do, we resort to a common mathematical trick and invert the problem:[1] we set $G(s) = 0$ and examine the resulting values of a_1, a_2. $G(s)$ is a *linear function* in a_1, a_2

$$G(s) = G_0(s) + a_1 G_1(s) + a_2 G_2(s)$$

where

$$G_0(s) = \frac{1}{4!}\left[(h-s)_+^4 - (-s)_+^4 - \frac{3}{2}h(h-s)_+^3 - \frac{19}{6}h(-s)_+^3 + \frac{5}{6}h(-h-s)_+^3 + \frac{h}{6}(-2h-s)_+^3\right]$$

$$G_1(s) = \frac{1}{4!}\left[(-s)_+^4 - (-h-s)_+^4 + \frac{h}{6}(h-s)_+^3 - \frac{13}{6}h(-s)_+^3 - \frac{13}{6}h(-h-s)_+^3 + \frac{h}{6}(-2h-s)_+^3\right]$$

$$G_2(s) = \frac{1}{4!}\left[(-s)_+^4 - (-2h-s)_+^4 - \frac{4}{3}h(-s)_+^3 - \frac{16}{3}(-h-s)_+^3 - \frac{4}{3}h(-2h-s)_+^3\right]$$

There are three intervals of s in which we must examine $G(s)$: $h \geq s \geq 0$, $0 \geq s \geq -h$, and $-h \geq s \geq -2h$. In the first interval

$$(h-s)^3\left(h - s - \frac{3}{2}h + a_1\frac{h}{6}\right) = 0$$

leads to

$$a_1 = 3 + \frac{6s}{h}$$

which describes a family of vertical lines in the a_1, a_2 plane. The line moves from $a_1 = 9$ to $a_1 = 3$ as s goes from h to 0 (Fig. 21.4.1).

[1] Carl Gustav Jacob Jacobi (1804–1851) said "Always invert."

In the second interval we have

$$\left[(h-s)^4 - s^4 - \frac{3}{2}h(h-s)^3 + \frac{19}{6}hs^3\right]$$
$$+ a_1\left[s^4 + \frac{h}{6}(h-s)^3 + \frac{13}{6}hs^3\right] + a_2\left(s^4 + \frac{4h}{3}s^3\right) = 0$$

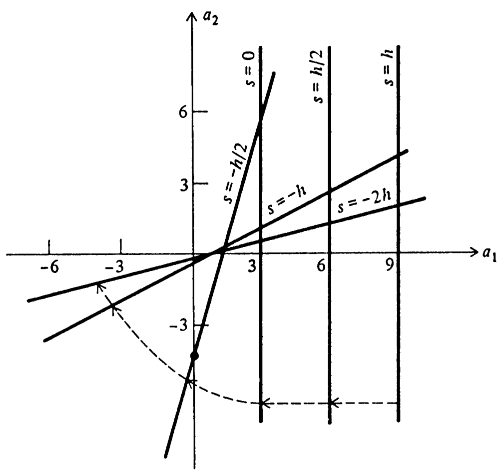

FIGURE 21.4.1
Region of constant sign for $G(s)$. (The lines are those along with $G(s) = 0$ for the value s marked on the line.)

This describes a line which continues to move to the left but gradually tilts until, at $s = -h$,

$$a_2 = \frac{a_1 - 1}{2}$$

This line passes through the point (1,0) with a slope of 1 : 2.

In the third interval we have the line rotating about the point (1,0) and ending up as

$$a_2 = \frac{a_1 - 1}{8}$$

Above and to the left of these lines (see Fig. 21.4.1), we have a $G(s)$ which has no zeros[1] and hence is of constant sign. Thus in this region we have the error term of the form

$$\frac{E_5 h^5}{5!} y^{(5)}(\theta)$$

Outside this region we do not have such an error term, and in view of future work, we shall not investigate it further.

In the region the error is measured by E_5 (Sec. 21.3). The "equi-error" lines are shown as slant lines in Fig. 21.4.2. These lines show that, other things

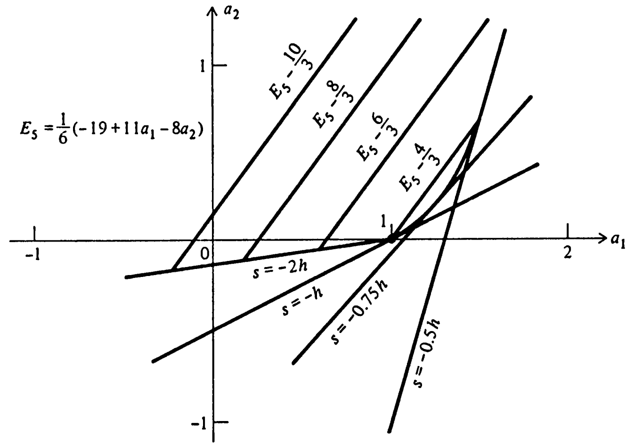

FIGURE 21.4.2
Equitruncation curves. (The value of E along the line gives the truncation error on that line.)

being the same, we should try to stay down toward the lower right where the error term is small. The slope of the lines, 11 : 8, indicates the "exchange ratio" between the two parameters; if we increase a_2 by 11 units and a_1 by 8, then the error is the same.

[1] There is also a region in the lower right for which $G(s)$ does not change sign, but both the size of the error term and the roundoff effects keep us from using such formulas, and we shall ignore it in the future. There is also a line for which $E_5 = 0$ and where we need a different $G(s)$ and we can ignore it for stability reasons.

21.5 STABILITY

We first examine the formulas by considering the trivial case

$$y' = 0$$

If this case does not behave favorably, then the more general case of

$$y' = f(x)$$

will also give trouble.

If we choose a point such as $a_1 = -1$ and $a_2 = 0$ in the a_1, a_2 plane, then the formula becomes

$$y_{n+1} = 2y_n - y_{n-1}$$

Now consider what happens to the solution $y = 0$ when a small error ε (which we can imagine coming from roundoff) is put in the solution at y_0. It is easy to compute Table 21.5.1. Thus the effect of the small error grows steadily.

As another example, choose $a_1 = 1$, and $a_2 = 1$. The equation is

$$y_{n+1} = -y_n + y_{n-1} + y_{n-2}$$

and we get Table 21.5.2.

This time the error oscillates as well as grows.

Table 21.5.1

$\vdots$
$y_{-2} = 0$
$y_{-1} = 0$
$y_0 = \varepsilon$
$y_1 = 2\varepsilon$
$y_2 = 3\varepsilon$
$y_3 = 4\varepsilon$
$\vdots$

Table 21.5.2

$\vdots$
$y_{-2} = 0$
$y_{-1} = 0$
$y_0 = \varepsilon$
$y_1 = -\varepsilon$
$y_2 = 2\varepsilon$
$y_3 = -2\varepsilon$
$y_4 = 3\varepsilon$
$y_5 = -3\varepsilon$
$\vdots$

These two examples indicate that we need to examine the phenomena further; clearly some points (a_1, a_2) are very poor choices.

The general form of the difference equation that we are investigating is a linear difference equation with constant coefficients, and we may apply the methods of Chap. 13. Since the values of y' are the integrand values which we compute independently of the y values, we consider the difference equation as being in the form

$$y_{n+1} - a_0 y_n - a_1 y_{n-1} - a_2 y_{n-2} = F_n$$

where F_n is a function of the derivative values y'_K which are the computed integrand samples.

We first solve the homogeneous equation

$$y_{n+1} - (1 - a_1 - a_2)y_n - a_1 y_{n-1} - a_2 y_{n-2} = 0$$

To solve this, we set $y_n = \rho^n$ to get the characteristic equation

$$\rho^3 - (1 - a_1 - a_2)\rho^2 - a_1\rho - a_2 = 0$$

or

$$(\rho - 1)[\rho^2 + (a_1 + a_2)\rho + a_2] = 0 \qquad a_2 \neq 0$$

Let the zeros of the quadratic factor be ρ_1 and ρ_2. The solution of the homogeneous equation is

$$y_n = C_1(\rho_1)^n + C_2(\rho_2)^n + C_3$$

provided that no two roots are equal. If two roots are equal, as they happen to be in the two examples above, then the solution needs to be changed. If either of the roots is greater in absolute value than 1, then the solution will grow similarly to a geometric progression as n increases (provided that the corresponding coefficient is not zero).

We therefore wish to explore the region of the a_1, a_2 plane for which $|\rho_i| < 1$. Again we invert the problem and determine the boundaries of the region. We first examine the boundary along which a root becomes equal to 1. To do this, we set $\rho = 1$ in the quadratic factor. This defines the line

$$1 + a_1 + 2a_2 = 0$$

$$a_2 = -\frac{1 + a_1}{2}$$

Along this line $\rho = 1, 1, a_2$ (the characteristic roots), and the solution has the form

$$y_n = C_1 + nC_2 + C_3(a_2)^n \qquad a_2 \neq 1$$

Next we examine where $\rho = -1$. To do this, we set $\rho = -1$ in the quadratic factor. This defines the line

$$a_1 = 1$$

Along this line $\rho = 1, -1, -a_2$.

Next we examine where the roots become complex. This curve is defined by

$$(a_1 + a_2)^2 - 4a_2 = 0$$

or

$$a_1 = \pm 2\sqrt{a_2} - a_2$$

which is shown in Fig. 21.5.1. Inside this parabola the roots are complex conjugates of each other and both have the modulus

$$|\rho| = \sqrt{a_2}$$

Thus the region in which $|\rho_i| < 1$ is bounded by the line

$$a_2 = 1$$

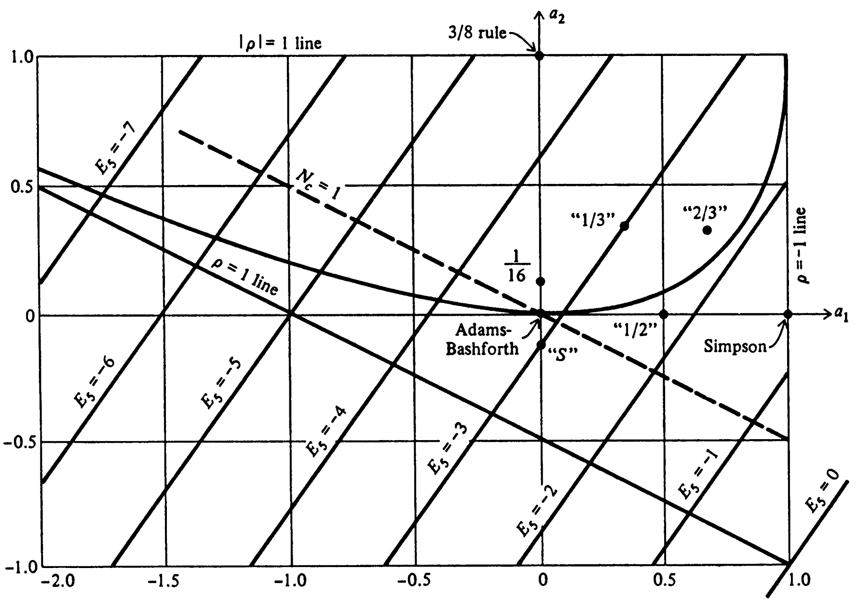

FIGURE 21.5.1
Stability region in a_1, a_2 plane.

which just cuts off the parabola where it is tangent to the other lines. These three straight lines in Fig. 21.5.1, then, define the triangle within which the characteristic roots must lie on order for the method of integration to have an error that does not grow as a geometric progression. Outside this region some $|\rho_i| > 1$. It is easy to see that the two illustrative examples in this section correspond to points which lie on the edge of the region and give double roots in the characteristic equation; the multiple roots, in turn, give rise to the linear growth. When the $|\rho_i| < 1$, we say that the method is *stable*; when $|\rho_i| = 1$, we say that it is *conditionally stable* (for simple roots only). The solution of the homogeneous equation is to be added to the solution of the complete equation. In solving the difference equation, any isolated roundoff will start the various solutions of the homogeneous equation, and unless the $|\rho_i| \leq 1$, there will be generated a solution that tends to infinity regardless of what F_n does.

The unwanted solutions of the difference equation occur because we have replaced a first-order differential equation by a third-order difference equation. The root $\rho = 1$ corresponds to the desired solution, and the other two ρ_i are the unwanted solutions. By requiring the $|\rho_i| \leq 1$ we are protected against their giving rise to large effects in the final answer. For some values of ρ such that $|\rho| = 1$, bounded solutions are possible.

PROBLEMS

21.5.1 Discuss the example where $a_1 = -1$ and $a_2 = 0$ and show that the computed table agrees with the theory when the proper equation is used.

21.5.2 Do the same as in Prob. 21.5.1 for $a_1 = 1$ and $a_2 = 1$.

21.5.3 Discuss the case of $a_1 = 2$ and $a_2 = 0$. Show that the theory and computation agree in giving a geometric growth in the error.

21.5.4 Discuss the behavior of the error as you cross each of the three bounding lines of stability region.

21.6 CORRELATED ROUNDOFF NOISE

The y_n values that we compute will, in general, have a roundoff noise level σ. At first thought the amplification factor of the roundoff noise would be taken as

$$N_a = (a_0^2 + a_1^2 + a_2^2)^{1/2}$$

However, because the y_{n+1} value is computed from the y_n, y_{n-1}, and y_{n-2} values, the noise in y_{n+1} is related to (correlated with) the noise in y_n, y_{n-1}, and y_{n-2}, and we need to give the matter more thought, although we still do not want N_a to be too big.

Suppose that we have a solution of

$$y' = 0$$

which is identically zero, and a small roundoff error occurs. The solution of the difference equation is

$$y_n = C_1(\rho_1)^n + C_2(\rho_2)^n + C_3$$

Assuming that the method is stable, $|p_i| < 1$, then

$$y_n \to C_3 \qquad \text{as } n \to \infty$$

Let us compute C_3 from the initial data

$$y_{-2} = 0$$
$$y_{-1} = 0$$
$$y_0 = \varepsilon$$

We have

$$\frac{C_1}{{\rho_1}^2} + \frac{C_2}{{\rho_2}^2} + C_3 = 0$$
$$\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2} + C_3 = 0$$
$$C_1 + C_2 + C_3 = \varepsilon$$

Solving for C_3, we get

$$C_3 = \frac{\varepsilon}{1 - (\rho_1 + \rho_2) + \rho_1\rho_2}$$

Using the quadratic factor in the characteristic equation, we have

$$\rho_1 + \rho_2 = -(a_1 + a_2)$$
$$\rho_1\rho_2 = a_2$$

Hence

$$C_3 = \frac{\varepsilon}{1 + a_1 + 2a_2} = N_c\,\varepsilon$$

where

$$N_c = \frac{1}{1 + a_1 + 2a_2}$$

Thus, if the roundoff error ε is not to end up greater than the starting value ε, then

$$|N_c| = |1 + a_1 + 2a_2|^{-1} \leq 1$$

and the smaller the better. Since we are mainly concerned with positive

$$1 + a_1 + 2a_2$$

we find that the condition for an isolated error ε not to grow is that (a_1, a_2) lie above the line

$$a_2 = -\frac{a_1}{2}$$

This line is shown in Fig. 21.5.1 as a dashed line.

PROBLEMS

21.6.1 The theory of roundoff leading to the result was developed for a single, isolated error. Each step will produce an error. How, if at all, does this change the argument?

21.7 SUMMARY

Table 21.7 lists a number of well-known methods which fall in the general class that we are investigating.

The choice $a_1 = 1$, $a_2 = 0$ is Simpson's method and lies on the edge of the stability region.

The choice $a_1 = 0$, $a_2 = 1$ is the three-eighths rule and also falls on the edge of the stability region.

The choice $a_1 = 0$, $a_2 = 0$ is labeled "Adams-Bashforth" because it is used in the Adams-Bashforth method of integrating differential equations (see Chap. 23).

All other methods of the general form are linear combinations of these three.

Figure 21.5.1 presents the relevant information. Each point in the a_1, a_2 plane corresponds to a method of integration. First, we want to choose the point inside the stability triangle, although we may venture onto the edge for some cases, such as Simpson's and the three-eighths-rule methods. Second, we want to stay toward the lower right to have a small truncation error (Fig. 21.4.2). Third, we wish to stay above the dashed line to keep the amplification of isolated error small.

Table 21.7 SOME METHODS FOR INTEGRATION

	Adams-Bashforth	Simpson	Three-eighths rule	1/3	1/2	2/3	−1/8*	1/16
a_0	1	0	0	1/3	1/2	0	9/8	15/16
a_1	0	1	0	1/3	1/2	2/3	0	0
a_2	0	0	1	1/3	0	1/3	−1/8	1/16
b_{-1}	9/24	1/3	3/8	13/36	17/48	25/72	3/8	18/48
b_0	19/24	4/3	9/8	39/36	51/48	91,72	6/8	39/48
b_1	−5/24	1/3	9/8	15/36	3/48	−43/72	−3/8	−6/48
b_2	1/24	0	3/8	5/36	1/48	9/72	0	3/48
E_5	−19/6	−4/3	−9/2	−3	−9/4	−43/18	−3	−39/12
Noise amplification N_c	1	1, 0, 1, 0, ...	1, 0, 0, ...	1/2	2/3	3/7	1.13	8/9
N_a	1	1	1	$1/\sqrt{3}$	$1/\sqrt{2}$	$\sqrt{5}/3$	$\sqrt{82}/8$	0.94

* Taken from R. Hamming, Stable Predictor-Corrector Methods for Ordinary Differential Equations, *J. Assoc. Computing Machinery*, vol. 6, no. 1, pp. 37–47, January, 1959.

Thus, except for the annoying tendency toward oscillation, Simpson's formula is one of the best of the general form examined (which does *not* include the first example of Sec. 21.2). The user should select his own method to suit the particular balance of factors in each problem, and no single best formula can be offered for all circumstances. The methods in Table 21.7 labeled 1/3, 1/2, and 2/3 are mixtures of the basic methods and are of interest in some applications.

PROBLEMS

21.7.1 Develop the theory of Secs. 21.3 to 21.7 (using only one parameter a_1) for the general form

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1}) + E_4 \frac{h^4 y^{(4)}(\theta)}{4!}$$

and compare with known results.

Ans.

	$a_1 = 0$*	Simpson	$a_1 = \frac{1}{5}$	
$a_0 = 1 - a_1$	1	0	$\frac{4}{5}$	$\rho_1 = -a_1$
$a_1 = a_1$	0	1	$\frac{1}{5}$	hence
$b_{-1} = (5 - a_1)/12$	$\frac{5}{12}$	$\frac{1}{3}$	$\frac{1}{10}$	stability
$b_0 = (8 + 8a_1)/12$	$\frac{8}{12}$	$\frac{4}{3}$	$\frac{8}{10}$	range
$b_1 = (-1 + 5a_1)/12$	$-\frac{1}{12}$	$\frac{1}{3}$	0	$-1 < a_1 < 1$
$E_4 = -1 + a_1$	-1	0	$-\frac{4}{5}$	E_4 not valid
Noise amplification				for
$N_c = 1/(1 + a_1)$	1	1, 0, 1, 0, ..., 1	$\frac{5}{6}$	$\frac{1}{5} < a_1 < 5$
N_a	1	1	$\sqrt{17}/5$	But $a_1 = 1$ higher order

* Southard and Yowell, *Math. Tables and Aids to Comp.*, vol. 6, pp. 253–254, 1952. [Clearly it is Simpson's half-formula see (15.6)]

21.7.2 Discuss the case of $a_1 = -1$.

21.8 SOME GENERAL REMARKS

The reader should be careful to note that only one particular class of formulas was investigated; other formulas are possible. For example, we could investigate formulas for which we used both integrand values (y') and derivatives of integrand values (y''). We would expect much better formulas in the sense of efficient computation, since frequently the derivative is readily calculated from various pieces of the computation of the integrand. However, this would involve more coding (and more mistakes) and for the occasional problem would probably be a poor strategy. For a recurring problem such formulas could save considerable machine time, and they should be investigated. See Prob. 21.8.1.

The powerful Gauss-type formulas have not been adapted to indefinite integral calculation because of the difficulty of fitting the peculiar spacing to consecutive steps.

The choice of E_5 as the error term has found wide favor in practice and represents a general compromise between different factors. Special situations, of course, require other choices.

The main purpose of this chapter was to introduce the idea of stability and the method of saving parameters to cope with it. We have not, therefore, gone into the involved subject of how to start an integration until enough back values are available to use the general formulas. The formulas of Sec. 21.2 can sometimes be used for this purpose.

PROBLEMS

21.8.1 Investigate the class of formulas

$$y_{n+1} = y_n + h(b_{-1}y'_{n+1} + b_0 y'_n) + h^2(c_{-1}y''_{n+1} + c_0 y''_n) + E_4 \frac{h^4 y^{(4)}(\theta)}{4!}$$

using b_{-1} as a parameter. Note the ease of starting.

Ans.

	General solution		A Milne method*			
b_{-1}	b_{-1}	1	$\frac{1}{2}$	0	$\frac{1}{3}$	$\frac{2}{3}$
b_0	$1 - b_{-1}$	0	$\frac{1}{2}$	1	$\frac{2}{3}$	$\frac{1}{3}$
c_{-1}	$(1 - 3b_{-1})/6$	$-\frac{2}{6}$	$-\frac{1}{12}$	$\frac{1}{6}$	0	$-\frac{1}{6}$
c_0	$(2 - 3b_{-1})/6$	$-\frac{1}{6}$	$\frac{1}{12}$	$\frac{2}{6}$	$\frac{1}{6}$	0
E_4	$-1 + 2b_{-1}$	1	0	-1	$-\frac{1}{3}$	$+\frac{1}{3}$
E_5	....	...	$\frac{1}{6}$			

*W. E. Milne, "Numerical Solution of Differential Equations," p. 76, John Wiley & Sons, Inc., New York, 1953, Dover, 1970.

The $G_4(s)$ changes sign for $1/3 < b_{-1} < 2/3$, but for the case $b_{-1} = 1/2$ the error is

$$\frac{E_5}{5!} h^5 y^{(5)}(\theta) = \frac{h^5}{720} y^{(5)}(\theta)$$

since

$$G(s) = \frac{(h-s)^2 s^2}{4!} \neq 0 \qquad 0 < s < h$$

and is particularly favorable.

21.8.2 Develop a method for starting an integration of the general form with due care to questions of accuracy.

21.9 EXPERIMENTAL VERIFICATION OF STABILITY

The reader should not become dismayed at this elaborate theoretical development. If a broad class of formulas is to be examined, some such approach is necessary; but if a *particular* formula is under consideration, then a simple experimental approach is often satisfactory.

Suppose that we are considering the formula

$$y_{n+1} = -y_n + \frac{3}{2} y_{n-1} + \frac{y_{n-2}}{2} + \frac{h}{48}(15y'_{n+1} + 85y'_n + 61y'_{n-1} + 7y'_{n-2})$$

It is easy to substitute $y = 1, x, x^2, x^3, x^4$ to see if they fit exactly.

With regard to stability, we simply try

$$y_{-2} = 0$$
$$y_{-1} = 0$$
$$y_0 = \varepsilon$$

and compute

$$y_1 = -\varepsilon$$
$$y_2 = \frac{5\varepsilon}{2}$$
$$y_3 = -\frac{7\varepsilon}{2}$$
$$\cdots\cdots\cdots\cdots$$

In this case it is clear that the method is unstable. A slight amount of care must be exercised to be sure that the experiments tried actually start all the possible terms (make each of the $C_i \neq 0$ at some time on some trial). Usually a single trial starts a linear combination of *all* the terms.

The direct experimental check for an unstable method is convincing and readily applied. The conclusion from negative evidence that a method is stable clearly requires some care, but often a particular formula is easy to handle.

The approach is illustrated in the next section.

*21.10 AN EXAMPLE OF A CONVOLUTION INTEGRAL WHICH ILLUSTRATES THE CONCEPT OF STABILITY

Oscillation with a geometric growth in amplitude is typical of stability troubles, and the following example shows how the various ideas can be applied in other situations.

The problem[1] was to compute $\Phi(t)$ when the data $\psi(t)$ were given at values of $\log t = -14.4(0.2)(0.8)$, and the equation connecting $\Phi(t)$ and $\psi(t)$ was

$$\int_0^t \Phi(t-\tau)\psi(\tau)\,d\tau = t \qquad (21.10.1)$$

or the equivalent

$$\int_0^t \Phi(\tau)\psi(t-\tau)\,d\tau = t \qquad (21.10.2)$$

The data $\psi(t)$ were monotone increasing.

A reasonable computing form was set up and tried. The results showed an oscillation with growing amplitude. The period of the oscillation was fixed, and the growth was more or less geometric, and so the conjecture was made that it was due to instability. An examination of the computing process was made with data which were uniformly $\psi(t) = 1$ but with a single perturbation; these data gave rise to oscillations (Sec. 21.9).

A new computing plan was sought. After some study, the following was adopted. We set

$$f(t) = \int_0^t \psi(\tau)\,d\tau$$

Thus $f(0) = 0$ and $f'(\tau) = \psi(\tau)$. Next we introduce notation t_i, $(i = 0, 1, \ldots, n)$ with $t_0 = 0$, $t_1 = 10^{-14.4}, \ldots, t_i = 10^{(-14.6+0.2i)}$ and *assumed* that $\psi(t)$ was constant between t_0 and t_1.

In order to calculate $f(t)$, we used the trapezoid rule for integration,

$$f(t_{n+1}) = f(t_n) + \tfrac{1}{2}[\psi(t_{n+1}) + \psi(t_n)](t_{n+1} - t_n)$$

since the data did not justify any more elaborate method.

We now had, from Eq. (21.10.2),

$$\begin{aligned} t_{n+1} &= \int_0^{t_{n+1}} \Phi(\tau)\psi(t_{n+1}-\tau)\,d\tau \\ &= \sum_{i=0}^{n} \int_{t_i}^{t_{i+1}} \Phi(\tau)\psi(t_{n+1}-\tau)\,d\tau \end{aligned}$$

In each of the integrals in the sum we made the approximation

$$\begin{aligned} \int_{t_i}^{t_{i+1}} \Phi(\tau)\psi(t_{n+1}-\tau)\,d\tau &= \Phi(t_{i+1/2}) \int_{t_i}^{t_{i+1}} f'(t_{n+1}-\tau)\,d\tau \\ &= -\Phi(t_{i+1/2})[f(t_{n+1}-t_{i+1}) - f(t_{n+1}-t_i)] \end{aligned}$$

[1] I. L. Hopkins and R. W. Hamming, On Creep and Relaxation, *J. Appl. Phys.*, vol. 28, pp. 906–909, August, 1957.

where $t_{i+1/2}$ is the midvalue $\frac{1}{2}(t_{i+1} + t_i)$. Thus

$$t_{n+1} = -\sum_{i=0}^{n-1} \Phi(t_{i+1/2})[f(t_{n+1} - t_{i+1}) - f(t_{n+1} - t_i)] + \Phi(t_{n+1/2})f(t_{n+1} - t_n)$$

Solving for $\Phi(t_{n+1/2})$, we got

$$\Phi(t_{n+1/2}) = \frac{t_{n+1} - \sum_{i=1}^{n-1} \Phi(t_{i+1/2})[f(t_{n+1} - t_i) - f(t_{n+1} - t_{i+1})]}{f(t_{n+1} - t_n)} \tag{21.10.3}$$

This provided a method for calculating each value $\Phi(t_{n+1/2})$ in turn, where the first value was given by

$$\Phi_{1/2} = \frac{t_1}{f(t_1)} \tag{21.10.4}$$

The reason that Eqs. (21.10.3) and (21.10.4) are successful, while a number of other ways of approximating the integral are not, is that an error in calculating a $\Phi(t)$ (due to faulty data or even just numerical roundoff of a product) does not grow in size in subsequent stages of the computation. To see this intuitively, we calculate the coefficient of $\Phi(t_{n-1/2})$ in the expression for $\Phi(t_{n+1/2})$. This coefficient is

$$-\frac{f(t_{n+1} - t_{n-1}) - f(t_{n+1} - t_n)}{f(t_{n+1} - t_n) - f(t_{n+1} - t_{n+1})}$$

[since $f(0) = 0$]. We apply the mean value theorem to both the numerator and the denominator separately.

$$\frac{\psi(t_{n+1} - \theta_1)(t_n - t_{n-1})}{\psi(t_{n+1} - \theta_2)(t_{n+1} - t_n)} = -\left[\frac{\psi(t_{n+1} - \theta_1)}{\psi(t_{n+1} - \theta_2)}\right]\left[\frac{t_n - t_{n-1}}{t_{n+1} - t_n}\right] \tag{21.10.5}$$

where $t_n > \theta_1 > t_{n-1}$ and $t_{n+1} > \theta_2 > t_n$. The terms in each bracket are less than 1 in magnitude, and so the error in $\Phi(t_{n-1/2})$ is multiplied by a number much less than 1 in size. Thus the effect of the error oscillates and dies out rapidly.

Since the answer was in the same form as the data and bore the same relationship [see Eqs. (21.10.1) and (21.10.2)], the answers were used as input data, and the original data reproduced well within experimental error. [We risked the fact that $\Phi(t)$ was *decreasing*, and hence the first factor in Eq. (21.10.5) was greater than 1; the second factor more than compensated for this for our data.]

In this example we see that the same phenomenon of instability occurs even though we have no fixed characteristic equation [compare Eq. (21.10.3)]. The general idea of feedback still applies. The above examination covered the feedback from only a single term; in practice it is usually necessary to make some examination to see that the additional feedback from other terms does not cause instability.

21.11 INSTABILITY IN ALGORITHMS

Iterative algorithms often show instability in much the same manner as illustrated in the last few sections.

For example, an iteration scheme may give a sequence of y_n values that oscillate and grow in size like a geometric progression. Thus we assume that

$$y_n = a + b\rho^n$$

where a is the "true" value. We write this for three consecutive values

$$\begin{aligned} y_{n-1} &= a + b\rho^{n-1} \\ y_n &= a + b\rho^n \\ y_{n+1} &= a + b\rho^{n+1} \end{aligned}$$

Transpose each a and divide the first equation by the second and the second by the third to get

$$\frac{y_{n-1} - a}{y_n - a} = \frac{1}{\rho} = \frac{y_n - a}{y_{n+1} - a}$$

Solving for a we have

$$a = \frac{y_{n+1}y_{n-1} - y_n^2}{y_{n+1} - 2y_n + y_{n-1}}$$

which reminds us of Shank's method [Eq. (12.7.3)].

Note that the process will improve convergence as well as cope with geometric divergence. After applying the method we iterate two more steps and reapply, etc., until the error is sufficiently small. If we have a function $f(t)$ being found by iteration we can apply the method at each point t_k.

22

INTRODUCTION TO DIFFERENTIAL EQUATIONS

22.1 THE SOURCE AND MEANING OF DIFFERENTIAL EQUATIONS

Our present views [10] of the fundamental laws of nature are often stated in the form of differential equations. For example, Newton's law

$$f = ma = m\frac{d^2x}{dt^2}$$

is a second-order differential equation whenever the force depends on position. As another example, consider a colony of bacteria growing in a medium which furnishes an adequate food supply. When the colony is studied in the large, it is natural to suppose that the rate of growth is proportional to the number $y(t)$ present at time t; that is,

$$y'(t) = ky(t)$$

Thus differential equations often provide our most primitive mathematical description of a natural phenomenon.

When possible, the differential equation is solved in closed form. Textbooks on differential equations often give the impression that most differential equations can be solved in closed form, but experience does not bear this out. Various analytical tools are available for obtaining approximate solutions, and these also should be examined before resorting to a numerical solution.

Given the differential equation

$$y' = x^2 - y^2$$

what do we mean by a solution? Setting aside the formal mathematical definition, we intuitively mean a curve $y = y(x)$ at each point (x, y) of which the slope $y'(x)$ of the curve is given by the differential equation. This is a *local property*. Solving an equation means extending this local property to a large region. There is, of course, not just a single curve which is a solution, but rather through *any* point (x_0, y_0) there is a solution of the differential equation (see the next section for exceptions).

22.2 THE DIRECTION FIELD

The ideas of the above paragraph can be made graphic. In the x, y plane we choose various points and compute the slope $x^2 - y^2$. At each of these points we draw a short line with the computed slope. These lines indicate the local direction of the solution, and by the use of a little imagination we can easily sketch in various solutions, provided the set of points through which we draw lines is sufficiently dense.

While the above method works, it is often easier to proceed in a somewhat different fashion. We ask for the curves along which all the short-line elements have the same slope: the *isoclines*, lines of the same slope. In the particular example that we are using, when we choose the slope k, we find the isoclines are hyperbolas:

$$x^2 - y^2 = k$$

The direction field using the isocline approach is shown in Fig. 22.2.1.

A number of things should be noted. If the solution curve has a maximum, or a minimum, then it must lie on the 0 isocline. Inflection points may similarly be located by setting $y'' = 0$:

$$\begin{aligned} y'' = 0 &= 2x - 2yy' \\ &= 2[x - y(x^2 - y^2)] \\ x &= \frac{1 \pm \sqrt{1 + 4y^4}}{2y} \end{aligned}$$

This curve is *not* shown in Fig. 22.2.1.

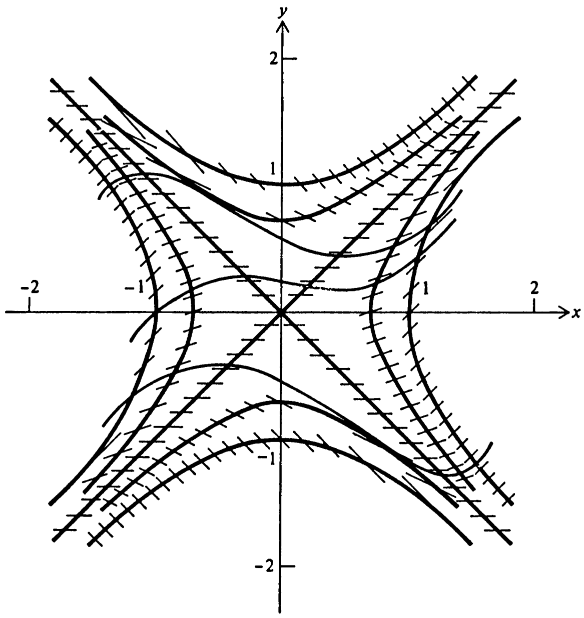

FIGURE 22.2.1
Direction field for $y' = x^2 - y^2$.

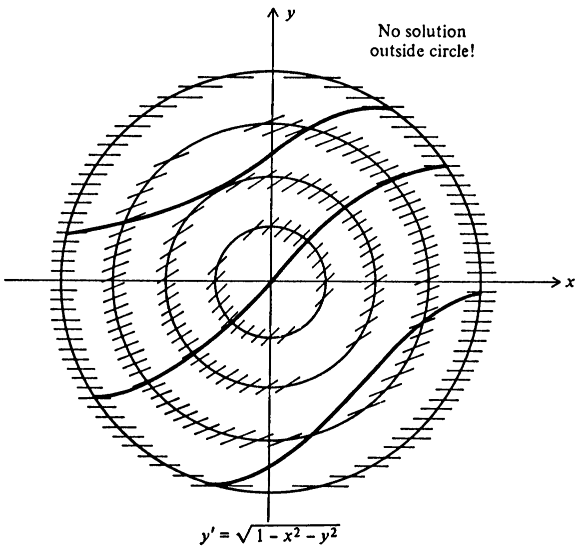

FIGURE 22.2.2

A direction-field picture can often shed a great deal of light on the nature of the problem. In the following case

$$y' = \sqrt{(1 - x^2 - y^2)}$$

Fig. 22.2.2 shows that the solution is confined to the circle and cannot get outside.

The direction field is a simple idea, but it should not be scorned. Frequently it shows the nature of the problem so that a sound approach may be made to the more accurate numerical solution. Several times in the author's experience, the direction-field sketch has answered the original problem, and on one occasion an accurate sketch prepared on a drawing board gave sufficient accuracy to answer the problem completely.

PROBLEMS

22.2.1 Draw the direction field and solutions for $y' = -\sin y$.

22.2.2 Draw the direction field and solutions for $y' = x - y^2$.

22.2.3 Draw the direction field and solutions for $y' = x^2 + y^2$.

22.3 THE NUMERICAL SOLUTION

If all that we want is one solution through a particular point, then it would be a waste of time to draw the entire direction field. Rather we would start at the point, compute the local direction, and then a little ahead of that compute the slopes at a few nearby points. And so we would proceed, always drawing only those slopes which are near where we expect to find the solution (Fig. 22.3.1).

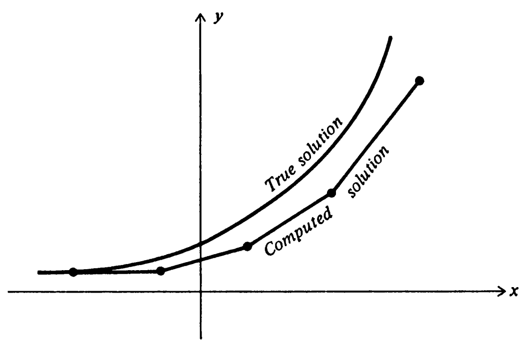

FIGURE 22.3.1
Crude numerical solution.

This process can be reduced to arithmetic. We take the starting values at point x_0, y_0, and using the slope y_0' at this point, we move forward a distance h to the next point

$$(x_0 + h, y_0 + hy_0') = (x_1, y_1)$$

Repeating this process, we go to the next point

$$(x_1 + h, y_1 + hy_1') = (x_2, y_2)$$

etc., and generate a numerical approximation to the solution of the differential equation.

For example, using the equation

$$y' = y^2 + 1 \qquad y_0 = 0 \qquad h = \tfrac{2}{10}$$

which has the known solution $y = \tan x$ for comparison purposes, we easily generate Table 22.3 which gradually drifts far from the solution. This is sometimes known as Euler's method and is clearly bad if large step sizes are used.

A simple look at Fig. 22.3.1 shows what is wrong with this method. When the true solution is concave upward, then the curve that we compute always lags below because it uses the slope from the past to get to the next point.

In view of this obvious fault we propose to look a bit ahead, *predict*, and then examine the slope at that point. Using this slope, we propose to go back and make a second, more *correct*, guess at how we should take the step. In more detail, we use the midpoint formula to *predict*

$$y_{n+1} = y_{n-1} + 2hy_n' \qquad (22.3.1)$$

Computing the slope at this point y_{n+1}' we then average the two end slopes as a more reasonable choice of slope and compute the *correct* value by the trapezoid formula

$$y_{n+1} = y_n + h\left(\frac{y_{n+1}' + y_n'}{2}\right) \qquad (22.3.2)$$

Table 22.3

x	$y_{n+1} = y_n + hy_n'$	$y' = y^2 + 1$	hy'	$y = \tan x$
0	0.000	1.00	0.200	0.000
0.2	0.200	1.04	0.208	0.203
0.4	0.408	1.17	0.234	0.423
0.6	0.642	1.41	0.282	0.684
0.8	0.724	1.52	0.304	1.030
1.0	1.028			1.557

The error term in the prediction is

$$\frac{h^3}{3} y'''(\theta_1) \qquad (22.3.3)$$

while the error term of the corrector is

$$-\frac{h^3}{12} y'''(\theta_2) \qquad (22.3.4)$$

The first question that occurs when we use this procedure is how to start. Our first prediction requires knowing one old point in addition to current slope. One answer is to use the differential equation and its derivatives to form a Taylor expansion

$$y_1 = y_0 + hy_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \cdots$$

about the starting point and from the series to compute the first point. Since the error term in each step depends on the third derivative, it is not necessary to extend the Taylor series beyond terms in h^3 or h^4.

This pattern of prediction and then correction has several nice properties. The difference between the predicted and the corrected values is, from Eqs. (22.3.3) and (22.3.4),

$$\frac{h^3}{12} [4y'''(\theta_1) + y'''(\theta_2)] \approx \frac{5h^3}{12} y'''(\theta)$$

If we assume that y''' does not change sign in the interval, then the two values are on opposite sides of the true value (for that single step). Thus the method provides a clue to the accuracy. If the error is too large, then the interval h can be shortened to $\bar{h} = h/2$ which multiplies the error by a factor of about $\frac{1}{8}$.

The reader should realize that we have been playing rather fast and loose with our symbols. We estimated a new value by Eq. (22.3.1) which is in error by Eq. (22.3.3). This value is used to compute the y'_{n+1}, and hence while we have written y'_{n+1} as if it were the true value, it is not; it is in error by approximately the second term in

$$f(x, y - \frac{h^3}{3} y'''(\theta)) \approx f(x, y) + \frac{\partial f(x, y)}{\partial y} \left[-\frac{h^3}{3} y'''(\theta) \right]$$

It is often proposed that we first predict and then correct, and if the two are not close enough, we then repeat the correction several times, using the latest estimate, until no appreciable change occurs. As a matter of principle, it is generally better to shorten the interval than to do repeated evaluations of the corrector formula.

22.4 AN EXAMPLE

Again consider the differential equation

$$y' = y^2 + 1$$
$$y(0) = 0 \qquad 0 \le x \le 1 \qquad (22.4.1)$$

(which was chosen *because* we know that the solution is $y = \tan x$, and we can use it for a check).

To get started, we construct the first few terms of the Taylor series by differentiating the differential equation

$$\begin{aligned} y' &= y^2 + 1 & y'(0) &= 1 \\ y'' &= 2yy' & y''(0) &= 0 \\ y''' &= 2yy'' + 2(y')^2 & y'''(0) &= 2 \\ y^{\text{iv}} &= 2y'y'' + 2yy''' + 4y'y'' & y^{\text{iv}}(0) &= 0 \end{aligned}$$

Hence

$$\begin{aligned} y(x) &= 0 + x \cdot 1 + \frac{x^2}{2} \cdot 0 + \frac{x^3}{3!} \cdot 2 + \frac{0 \cdot x^4}{4!} + \cdots \\ &= x + \frac{x^3}{3} + \cdots \end{aligned}$$

We shall use the crude spacing $h = \Delta x = 0.2$. We have

$$y_1 = y(0.2) = 0.2 + \frac{0.008}{3} = 0.203$$

as the first point, and (using a slide rule)

$$y_1' = y'(0.2) = (0.203)^2 + 1 = 1.0412$$

We are now ready to start using the method. We *predict* the next y value, where we now write p for clarity

$$p_2 = y_0 + 2hy_1' = 0 + 0.4(1.0412) = 0.4165$$

and evaluate, using the differential equation (22.4.1):

$$p_2' = 1.173$$

Next we compute the corrector value, where we now write c for clarity:

$$c_2 = 0.203 + 0.1(1.0412 + 1.173) = 0.424$$

Now we know that p_2 is in error by

$$\frac{h^3}{3} y'''(\theta_1)$$

and c_2 is in error by

$$-\frac{h^3}{12} y'''(\theta_2)$$

If y''' is approximately constant in the interval, then

$$p - c = \frac{5h^3}{12} y'''$$

Thus the error in the corrector value is approximately

$$-\tfrac{1}{5}(p - c)$$

Hence we *add*

$$\tfrac{1}{5}(p - c) = -0.002$$

to obtain the final value

$$y = c + \tfrac{1}{5}(p - c) = 0.422$$

We are now ready to repeat the cycle until we reach $x = 1$. (See Table 22.4.)

As a check we find that we have computed $y(1) = 1.546$ while the true value is 1.557. The error 0.011 represents a relative error

$$\frac{0.011}{1.557} \times 100 = \frac{7}{10} \text{ percent}$$

which is fairly good, considering the spacing and the use of a slide rule. This is known as the modified Euler process for integrating an ordinary differential equation.

PROBLEMS

22.4.1 Solve numerically $y' = y^2$, $(0) = 1$, $h = 0.2$ for $0 \le x \le 1$.
22.4.2 Solve numerically $y' = y^2 - \sin^2 x$, $y(0) = 0$, $h = \pi/6$ for $0 \le x \le 2\pi$.

Table 22.4 NUMERICAL SOLUTION OF $y' = y^2 + 1, y(0) = 0$

x	y	y'	p	p'	c	$p - c$	$\frac{p-c}{5}$
0	0	1					
0.2	0.203	1.0412					
0.4	0.422	1.178	0.41648	1.173	0.424	−0.008	−0.002
0.6	0.683	1.466	0.674	1.455	0.685	−0.011	−0.002
0.8	1.027	2.047	1.008	2.016	1.031	−0.023	−0.004
1.0	1.546		1.502	3.256	1.557	−0.055	−0.011
Correct value	1.557						
Error	0.011						

22.5 STABILITY OF THE PREDICTOR ALONE

It is sometimes proposed that the predictor formula

$$y_{n+1} = y_{n-1} + 2hy'_n \qquad (22.5.1)$$

be used with no corrector following. Let us examine the stability of such a proposal before we examine the whole predictor-corrector process.

Let $z(x)$ be the true solution of the equation

$$z' = f(x, z) \qquad (22.5.2)$$

and let y_n be the computed solution at $x = x_n$. If the computed solution y_n is put in the differential equation, it will fit exactly, since the computed y' value was found from the equation. Thus we have (neglecting roundoff errors)

$$y'_n = f(x, y_n) \qquad (22.5.3)$$

Set

$$\varepsilon_n = z_n - y_n$$

Then subtracting Eq. (22.5.3) from (Eq. (22.5.2), we get

$$\begin{aligned} \varepsilon'_n = z'_n - y'_n &= f(x, z_n) - f(x, y_n) \qquad (22.5.4) \\ &= \frac{\partial f(x, \theta)}{\partial y} \varepsilon_n = A\varepsilon_n \end{aligned}$$

where A is defined to be $A = \partial f/\partial y$ and θ lies between y_n and z_n.

When we substitute the true solution in the difference equation of the predictor, we get an error $e(x)$, since the true solution does not generally satisfy the difference equation

$$z_{n+1} = z_{n-1} + 2hz'_n + e_n \qquad (22.5.5)$$

This time subtracting the two difference equations, Eq. (22.5.1) from Eq. (22.5.5), we get

$$\varepsilon_{n+1} = \varepsilon_{n-1} + 2h\varepsilon'_n + e_n \qquad (22.5.6)$$

We are concerned with *estimating* the *local growth* of the error ε_n. For this purpose we may assume that e_n and $\partial f/\partial y = A$ are all constants. Unless they vary slowly, the step size of the integration is too large. Under these assumptions, we may put Eq. (22.5.4) in Eq. (22.5.6) to get

$$\varepsilon_{n+1} = \varepsilon_{n-1} + 2hA\varepsilon_n + e_n$$

or

$$\varepsilon_{n+1} - 2Ah\varepsilon_n - \varepsilon_{n-1} = e_n \qquad (22.5.7)$$

which is a linear difference equation with constant coefficients.

The characteristic equation of this difference equation is

$$\rho^2 - 2Ah\rho - 1 = 0 \qquad (22.5.8)$$

Hence the general solution is

$$\varepsilon_n = C_1(\rho_1)^n + C_2(\rho_2)^n \qquad (22.5.9)$$

where

$$\begin{aligned} \rho_1 &= Ah + \sqrt{A^2h^2 + 1} \\ \rho_2 &= Ah - \sqrt{A^2h^2 + 1} \end{aligned} \qquad (22.5.10)$$

If $A = \partial f/\partial y > 0$, then $\rho_1 > 1$; while if $A < 0$, then $|\rho_2| > 1$. In either case one of the two terms $\rho_1{}^n$ or $\rho_2{}^n$ becomes large as $n \to \infty$. Hence the method is said to be unstable.

Thus we see that this predictor formula, used by itself alone, gives rise to a solution that is not stable since one of the zeros of the characteristic equation (22.5.8) will be $|\rho_i| > 1$. But see Sec. 22.6.

PROBLEMS

22.5.1 Try solving numerically $y' = -y$, $y(0) = 1$, $h = 0.1$, for $0 \le x \le 1$, using the predictor only, and compare the results with theory.

22.6 STABILITY OF THE CORRECTOR

The proposal that we iterate the corrector until no change occurs can be investigated by the same method we used on the predictor. The corrector formula

$$y_{n+1} = y_n + \frac{h}{2}(y'_{n+1} + y'_n) \qquad (22.6.1)$$

leads, by exactly the same argument, to the characteristic equation

$$\left(1 - \frac{Ah}{2}\right)\rho - \left(1 + \frac{Ah}{2}\right) = 0$$

$$\rho = \frac{1 + Ah/2}{1 - Ah/2} = 1 + Ah + \frac{(Ah)^2}{2} + \frac{(Ah)^3}{4} + \cdots \qquad (22.6.2)$$

provided that $|Ah/2| < 1$.

We first examine $Ah/2$ by examining the iteration process used on the corrector. Let c_i be the ith iteration. We compute the change in the iteration each step by writing

$$\begin{aligned} c_{i+1} - c_i &= \frac{h}{2}\,[f(x, c_i) - f(x, c_{i-1})] \\ &= \frac{h}{2}\frac{\partial f}{\partial c_i}(c_i - c_{i-1}) \\ &\approx \frac{Ah}{2}(c_i - c_{i-1}) \end{aligned}$$

Evidently the process will converge if and only if the *convergence factor*

$$\left|\frac{Ah}{2}\right| < 1 \qquad (22.6.3)$$

Returning to Eq. (22.6.2), we note that ρ is given by the first three terms of e^{Ah} plus a slight error in the fourth term.

In the simple equation

$$y' = Ay$$

where to be specific we assume $A > 0$, we have

$$\frac{\partial f}{\partial y} = A$$

and the true solution at x_n is

$$y_n = Ce^{Ahn}$$

which is the solution by Eq. (22.6.2) (if terms in h^3 are neglected).

The criterion that a characteristic root $|\rho_i| \approx |e^{Ah}| < 1$ means instability is thus seen to be misleading. Evidently, if the solution grows as a geometric progression, errors that grow at the same rate or less will not necessarily ruin the solution. On the other hand, if $A < 0$, then the errors must die out at least as rapidly as e^{Ahn}, or they will dominate the solution being computed. The fact that the errors are bounded is no help in the situation of a decreasing solution. Loosely speaking, the number of adjacent solution curves that we cross is the significant factor when we make an error.

This suggests that what we want for differential equations is the criterion of *relative stability*, which is defined as the rate of growth of the error *relative* to the growth of the solution. When the relative stability is less than 1 in size, then the noise due to an isolated roundoff, or other, error will not grow more rapidly than the solution.

In view of this new criterion of relative stability, it is necessary to reexamine the discussion in Sec. 22.5 of the use of the predictor alone. If we assume that $|Ah| < 1$, then the characteristic roots ρ_i can be expanded as

$$\rho_1 = Ah + \sqrt{1 + (Ah)^2} = Ah + 1 + \frac{(Ah)^2}{2} - \frac{1}{8}(Ah)^4 + \cdots$$

$$\rho_2 = Ah - \sqrt{1 + (Ah)^2} = Ah - [1 + \frac{(Ah)^2}{2} - \frac{1}{8}(Ah)^4 + \cdots]$$

or

$$\rho_1 = 1 + Ah + \frac{(Ah)^2}{2} + \cdots \approx e^{Ah}$$

$$\rho_2 = -\left[1 - Ah + \frac{(Ah)^2}{2} - \cdots\right] \approx -e^{-Ah} \qquad (22.6.4)$$

The solution itself grows as e^{Ah}. From this we see that if $A > 0$, then the repeated use of the predictor can be expected to give reasonable results; but if $A < 0$, then $(\rho_2)^n$ will grow in size and oscillate, while the true solution decreases. Hence the method of repeated use of the predictor is relatively unstable only when $A < 0$. (See Prob. 22.5.1.)

Returning to the iterated corrector process, we find that the characteristic equation has only one root, Eq. (22.6.2), and the solution behaves exactly as this root indicates; thus the iterated corrector method is relatively stable.

PROBLEMS

22.6.1 Show that with the use of only the predictor, the result of numerically integrating $y' = y$, $y(0) = 1$, $h = 0.2$ for $0 \leq x \leq 1$ agrees with theory.

22.7 SOME GENERAL REMARKS

We have now given four examples of stability analysis, the first applied to indefinite integrals, the second to a particular convolution integral, and the last two to differential equations. Note that stability for integrals and for differential equations is not the same; for differential equations there are feedback paths from the derivatives which are not present in the case of integrals. The necessity for stability analysis arises whenever old values are used to compute new values and thus produce feedback of errors. There is no simple rule to apply in all cases, but it should be evident that the subject of stability is closely connected with feedback, a subject on which many books have been written. We have also indicated that *for differential equations the concept of stability is misleading, and relative stability is a more reasonable criterion.*

The stability analysis that we did on the simple integration system was incomplete. We analyzed the predictor and the corrector separately, neglected any cross-effects, and ignored the final mop-up of the error. It is possible to write the complete system of difference equations and construct the corresponding characteristic equation, but the results are very involved. Rather, we prefer to make a few remarks about how they interact. The instability of the predictor has very little effect on the relative stability of the result after the corrector is applied once. The final mop-up also causes little trouble in regard to relative stability.

It may occur to the reader that the same argument that we used to mop up the error of the corrected value could be used to modify the predicted value and thus reduce the error in the evaluation of the predicted derivative which enters into the final value. To do this, we could add to the predicted value $-\frac{4}{5}(p_n - c_n)$ of the last cycle. If there is a tendency to oscillate, this will emphasize it. On the other hand, there is no doubt that if instability and roundoff are not troubling the computation, such a term has a good effect on the accuracy of the whole computation.

PROBLEMS

22.7.1 Develop the formulas for the integration system of predict, modify, correct, and final value.

22.8 SYSTEMS OF EQUATIONS

A single differential equation of nth order

$$y^{(n)} = f(x, y, y', \ldots, y^{(n-1)})$$

can be reduced to a system of n first-order equations by a simple change in notation. Write

$$y = y_0 \qquad y' = y_1 \qquad y'' = y_1' = y_2$$
$$y''' = y_2' = y_3, \ldots, y^{(n-1)} = y_{n-1}$$

and we have the equivalent system

$$\begin{aligned} y_0' &= y_1 \\ y_1' &= y_2 \\ y_2' &= y_3 \\ &\cdots\cdots\cdots \\ y_{n-1}' &= f(x, y_0, y_1, y_2, \ldots, y_{n-1}) \end{aligned}$$

In order to apply the method of integration to a system, we simply apply the operations to the separate equations in parallel. The cross connections between the equations are taken care of when the derivatives are computed; otherwise the equations are treated separately, but in parallel. However, see Chap. 24 for other methods.

The direction-field method is generally applicable to a single first-order equation only, but in the case of two first-order equations in which the independent variable x does not occur,

$$y' = f(y,z)$$
$$z' = g(y,z)$$

dividing one equation by the other produces a first-order equation without x. The direction field for this equation gives y versus z, and sometimes it is sufficient to indicate the trajectory of y versus z without specifying the x for which the point y, z on the curve occurs.

23

A GENERAL THEORY OF PREDICTOR-CORRECTOR METHODS

23.1 INTRODUCTION

Chapters 21 and 22 introduced ideas and techniques which we now use to develop a general theory of predictor-corrector methods for numerically integrating ordinary differential equations. This theory includes the Milne and Adams-Bashforth methods as special cases. Thus we are in a position to compare these and other widely used methods within the framework of a single theory.

Predictor-corrector methods for integrating ordinary differential equations are widely used because of the following advantages:

1. The difference between the predicted and corrected values provides one measure of the error being made at each step and hence can be used to control the step size employed in the integration.
2. Only one or two evaluations of the derivatives need to be computed at each step (as compared with four for the Runge-Kutta method of Chap. 24), and on high-order systems this can save considerable computing effort.
3. It is easy to catch many kinds of machine failures.

Some disadvantages are that the methods are complex to program and are not self-starting. A method for starting will be discussed in Chap. 24.

There are three main sources of trouble in predictor-corrector methods for integrating ordinary differential equations. They are:

1. Truncation errors that arise from the finite approximations for the derivatives
2. Propagation errors (instability) that arise from solutions of the approximate difference equations that do not correspond to solutions of the differential equations
3. Amplification of roundoff errors due to certain combinations of coefficients in the finite difference formulas

The plan of this chapter is to examine a broad class of formulas in a systematic fashion and indicate how the choice of a particular formula is a compromise between conflicting desires. Since the corrector formula plays the more important role in the integration of ordinary differential equations, it is natural to examine the corrector first and more intensively.

After examining the corrector, we take up the corresponding problem for the predictor. With families of both predictors and correctors available, we are then in a position to examine the problem of designing a system for integrating ordinary differential equations. We conclude with a discussion of some experimental results.

The methods examined have a truncation error of order h^5; experience shows that these are the most widely used. Chapter 22 developed a predictor-corrector method of order h^3, and the intermediate theory of order h^4 is left as exercises.

The third-order method of Chap. 22 had no free parameters; the fourth-order methods treated in the exercises introduce two new coefficients, one of which is used to increase the order of the truncation error and one for stability; the fifth-order methods in the text have still another two coefficients which are again used, one for truncation and one for stability, and so on for higher-order methods.

In many applications the truncation error of the finite approximation to the derivatives of a differential equation, or to a system of differential equations, greatly exceeds the errors due to roundoff at each step. However, in building special-purpose computers that must operate in real time, there is great pressure to keep the length of the numbers, and hence the basic level of accuracy, as low as possible. In such situations the truncation error may be about the same size as the roundoff error, and the latter cannot be ignored in designing the methods for integrating the ordinary differential equations of the problem. It also happens with increasing frequency these days that in some problems, such as trajectories to the moon and planets, eight- and even ten-decimal-place accuracy can produce significant roundoff errors, and some methods described here are then useful. However, whole books have been written on the topic of numerical integration of ordinary differential equations, and we can cover only a small part of the topic.

23.2 TRUNCATION ERROR

The most general linear corrector formula that uses the function and the first-derivative information at the last three points of the solution, plus an estimate of the derivative at the point being computed, is

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1} y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2}) + E_5 \frac{h^5 y^{(5)}}{5!}(\theta)$$

Other linear forms could be used, but this one includes many of the common methods and is the same as we used in Chap. 21 when examining indefinite integrals. The change from the treatment in Chap. 21 is that now we have feedback through the derivative terms, as well as from old solution values; hence the investigation of the stability is more complex.

The formula as written implies that it is exact for polynomials through degree 4; that is, it is exact for $y = 1, x, x^2, x^3, x^4$. Using these five conditions and taking a_1 and a_2 as parameters, we get, as before in Sec. 21.3,

$$\begin{aligned} a_0 &= 1 - a_1 - a_2 & b_0 &= \tfrac{1}{24}(19 + 13a_1 + 8a_2) \\ a_1 &= a_1 & b_1 &= \tfrac{1}{24}(-5 + 13a_1 + 32a_2) \\ a_2 &= a_2 & b_2 &= \tfrac{1}{24}(1 - a_1 + 8a_2) \\ b_{-1} &= \tfrac{1}{24}(9 - a_1) & E_5 &= \tfrac{1}{6}(-19 + 11a_1 - 8a_2) \end{aligned}$$

Five of the seven coefficients are used to reduce the truncation error; the other two will be used to reduce the instability and roundoff errors rather than to reduce the truncation error still further. As in Chap. 21, the difference of the predictor and corrector values will be used both to provide an estimate of the error and to "mop up" the leading error terms.

The usual method of finding the error term is to use the Taylor expansion (Sec. 16.2),

$$y(x) = y(A) + (x - A)y'(A) + \frac{(x - A)^2}{2!} y''(A) + \cdots + \frac{(x - A)^{m-1}}{(m - 1)!} y^{(m-1)}(A) + \frac{1}{(m - 1)!} \int_A^x y^{(m)}(s)(x - s)^{m-1}\, ds$$

with $A = -2h$ and $m = 5$. Following the steps in Chap. 21, we obtain $(B = h)$

$$\text{Error} = R(y) = \int_{-2h}^{h} y^{(5)}(s)G(s)\,ds$$

where $G(s)$ is the *influence function*

$$G(s) = (\text{LHS} - \text{RHS})\frac{(x-s)_+^{m-1}}{(m-1)!}$$

$$4!\,G(s) = (h-s)_+^4 - a_0(-s)_+^4 - a_1(-h-s)_+^4 - a_2(-2h-s)_+^4$$
$$- 4h[b_{-1}(h-s)_+^3 + b_0(-s)_+^3 + b_1(-h-s)_+^3 + b_2(-2h-s)_+^3]$$

and (see Sec. 16.4)

$$(a-s)_+^k = \begin{cases}(a-s)^k & \text{if } a-s \ge 0 \\ 0 & \text{if } a-s \le 0\end{cases} \qquad k > 0$$

If $G(s)$ is of constant sign, then there exists a θ $(-2h < \theta < h)$ such that the error can be written as

$$\text{Error} = y^{(5)}(\theta)\int_{-2h}^{h} G(s)\,ds$$

If $G(s)$ is not of constant sign, then there will be functions $f(s)$ for which the error does not have this general form (Sec. 16.7).

The investigation in Sec. 21.4 produced the region of constant sign for $G(s)$ (shown in Figs. 21.4.1 and 21.4.2 of Sec. 21.4), in which the main interest lies and for which the lines of equal truncation error are shown.

23.3 STABILITY

Following the method of Sec. 21.5, we let $z = z(x)$ be the true solution, which means that z satisfies both

$$z' = f(x, z) \qquad (23.3.1)$$

and

$$z_{n+1} = a_0 z_n + a_1 z_{n+1} + a_2 z_{n-2}$$
$$+ h(b_{-1}z'_{n+1} + b_0 z'_n + b_1 z'_{n-1} + b_2 z'_{n-2}) + e_n \qquad (23.3.2)$$

and $y = y_n$ be the calculated solution

$$y'_n = f(x_n, y_n) \qquad (23.3.3)$$

$$y_{n+1} = a_0 y_n + a_1 y_{n-1} + a_2 y_{n-2} + h(b_{-1}y'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2}) \qquad (23.3.4)$$

We set

$$\varepsilon_n = z_n - y_n$$

and subtract Eqs. (23.3.2) and (23.3.4) to get

$$\varepsilon_{n+1} = a_0 \varepsilon_n + a_1\varepsilon_{n-1} + a_2 \varepsilon_{n-2} + h(b_{-1}\varepsilon'_{n+1} + b_0 \varepsilon'_n + b_1\varepsilon'_{n-1} + b_2 \varepsilon'_{n-2}) + e_n \tag{23.3.5}$$

while Eqs. (23.3.1) and (23.3.3) give, using the mean value theorem,

$$\begin{aligned}\varepsilon'_n &= f(x_n, z_n) - f(x_n, y_n) \\ &= \frac{\partial f(x_n, \theta)}{\partial y}\,\varepsilon_n = A\varepsilon_n\end{aligned} \tag{23.3.6}$$

where θ is the usual mean value. Putting Eq. (23.3.6) in Eq. (23.3.5), we get

$$(1 - b_{-1}Ah)\varepsilon_{n+1} = (a_0 + b_0 Ah)\varepsilon_n + (a_1 + b_1Ah)\varepsilon_{n-1} + (a_2 + b_2 Ah)\varepsilon_{n-2} + e_n \tag{23.3.7}$$

which is a linear difference equation.

In studying the growth of the error ε_n, it is reasonable to assume that e_n and $\partial f/\partial z = A$ are both constants; in practice they vary slowly from step to step. Hence we assume that Eq. (23.3.7) is a linear difference equation with constant coefficients (locally).

In Eq. (23.3.7) only the quantity Ah occurs, not h or A separately. While Ah can vary widely, it is necessary to estimate reasonable values that may be expected. To gain a little perspective, consider the equations

$$y' = \pm Ay \qquad y(0) = 1$$

The truncation error at each step will be taken as $h^5A^5/40$ ($\frac{1}{40}$ is a typical value for $E_5/5!$; see Table 21.7). This leads to Table 23.3. The table shows that if the truncation error is to be reasonably small for these equations, then $|A| \leq 5/10$ and for accurate solutions $|Ah| \leq 2/10$.

Table 23.3 TRUNCATION ERROR AS A FUNCTION OF $|Ah|$

$\lvert Ah\rvert$	$\frac{h^5y^{(5)}}{40}$
0.1	2.50×10^{-7}
0.2	8.00×10^{-6}
0.3	6.07×10^{-5}
0.4	2.56×10^{-4}
0.5	7.88×10^{-4}

Of course other equations could have different patterns of higher derivatives (probably growing more rapidly than simple powers), and one cannot say anything very specific about what $|Ah|$ will be in comparison with $E_5 h^5 y^{(5)}/5!$. Nevertheless, we shall adopt the attitude of $|Ah| < 0.4$ for most equations of interest.

Figure 21.5.1 shows the stability region in case $Ah = 0$. When $Ah \neq 0$, the corresponding homogeneous difference equation leads to the characteristic equation

$$Ah = \frac{\rho^3 - a_0\rho^2 - a_1\rho - a_2}{b_{-1}\rho^3 + b_0\rho^2 + b_1\rho + b_2} \tag{23.3.8}$$

This equation is more complex than the corresponding equation in Chap. 21 because of the extra feedback from the derivative terms. The solution is

$$\varepsilon_n = C_1(\rho_1)^n + C_2(\rho_2)^n + C_3(\rho_3)^n \tag{23.3.9}$$

The solution of the original differential equation (23.3.1) tends to grow as

$$y_n = Ce^{Ahn}$$

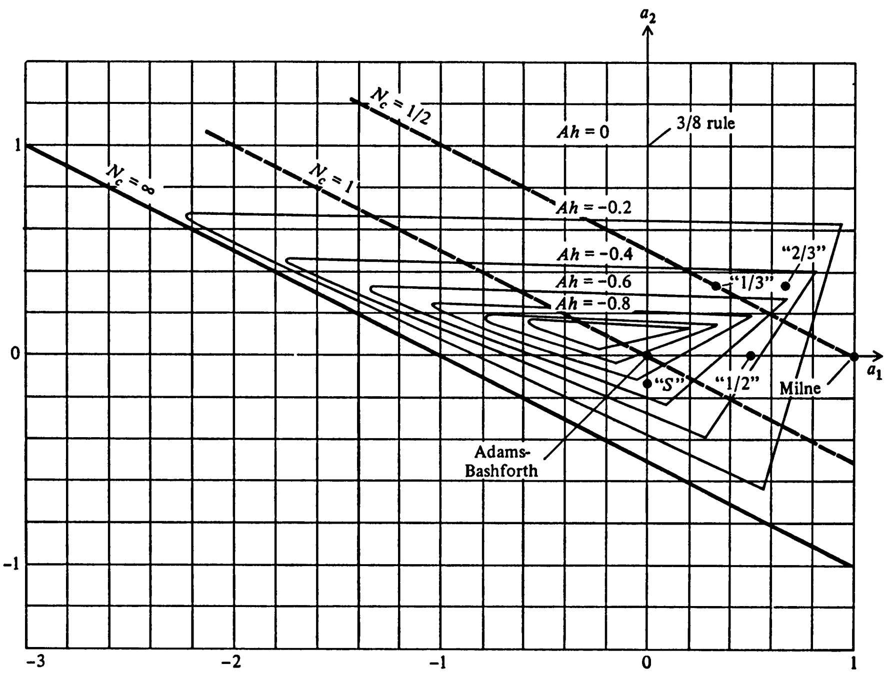

FIGURE 23.3.1
Stability region.

and generally it is not the growth of the error in itself but the growth *relative* to the local growth of the solution which concerns us. Thus it is not stability

$$|\rho_i| < 1$$

that we want, but rather *relative stability*

$$|\rho e^{-Ah}| \leq 1 \qquad (23.3.10)$$

One of the roots, which we shall designate as ρ_3, behaves as

$$\rho_3 = 1 + Ah + \frac{(Ah)^2}{2} + \frac{(Ah)^3}{3!} + \frac{(Ah)^4}{4!} + k\frac{(Ah)^5}{5!} \simeq e^{Ah} \qquad (23.3.11)$$

for small Ah. Thus

$$\rho_3 e^{-Ah} \simeq 1$$

and ρ_1, ρ_2 are the roots which may give relative instability trouble.

The triangle along which, for $Ah = 0$, the roots take on the values

$$|\rho e^{-Ah}| = |\rho| = 1$$

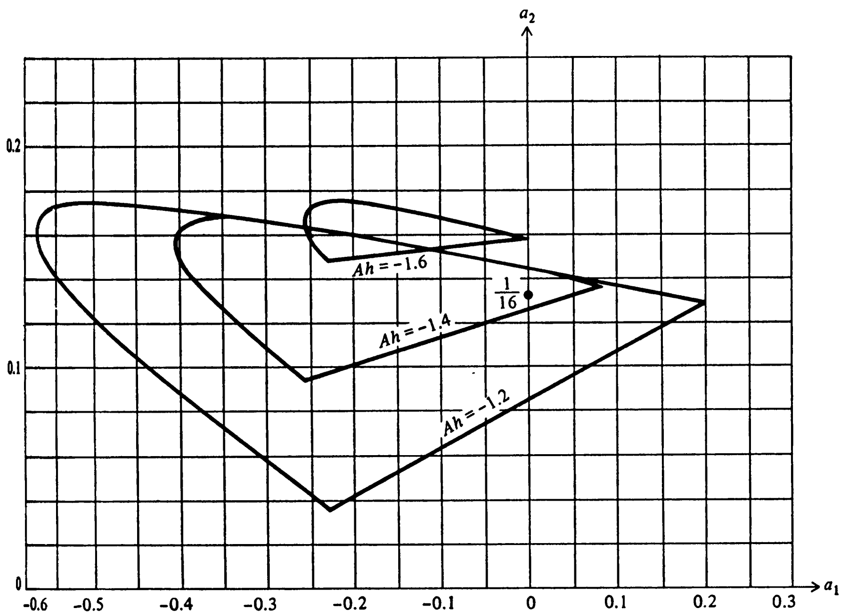

FIGURE 23.3.2
Stability region.

becomes, for nonzero Ah, somewhat more complex. For $\rho = -e^{Ah}$ we obtain a linear equation

$$\alpha a_2 + \beta a_1 + \gamma = 0 \quad (23.3.12)$$

where α, β, and γ are combinations of Ah, e^{Ah}, e^{2Ah}, and e^{3Ah}. But the other two sides of the triangle become a hyperbola. In Figs. 23.3.1 and 23.3.2 we show for negative values of Ah one branch of the hyperbola until it is cut off by the line of Eq. (23.3.12). While values inside the other branch would be stable, in the intervening gap there would be unstable values; and so they do not seem to be useful except possibly for special situations. These charts, Figs. 23.3.1 and 23.3.2, show stability regions for the roots ρ_1 and ρ_2; for large values of $|Ah|$ the root ρ_3 is far from its required value of e^{Ah}, and the truncation error cannot be neglected.

PROBLEMS

23.3.1 Plot the line of Eq. (23.3.12) for $Ah > 0$, and discuss the region of stability.

23.3.2 Using Prob. 21.7.1, develop the corresponding theory for differential equations. Draw the stability region as a function of a_1 and Ah.

23.4 ROUNDOFF NOISE

The theory of the propagation of roundoff noise for a differential equation is very much like that for indefinite integrals treated in Sec. 21.6. For $Ah = 0$ the line

$$a_2 = -\frac{a_1}{2}$$

is the dividing line between the region of growth and the region of decay of a random error. Inside the stability region the noise amplification is given by (see Sec. 21.6)

$$N_c = \frac{1}{1 + a_1 + 2a_2}$$

When $Ah \neq 0$, the division between the two regions is more complex, but it is not worth exploring it here; we merely need to recall that it is better to be above the line than below it.

To control the uncorrelated noise, it is well to keep

$$N_a = (a_0{}^2 + a_1{}^2 + a_2{}^2)^{1/2}$$

reasonably small.

On machines that "chop," that is, drop the digits rather than round properly, there is a second kind of trouble with roundoff. If, for the moment, we regard the various numbers as if they were mass, then chopping a positive number decreases the mass, while chopping a negative number increases the mass. In most computations we can reasonably expect some balance between the positive and negative numbers. But for a differential equation in some interval there well may be an excess of one effect or another, and in the next interval, since the signs of the numbers involved tend to change infrequently, there will be an effect *in the same* direction. Thus step after step we will find that chopping pushes us in the same direction, and the effect accumulates linearly, rather than in a customary way such as the square root of the number of operations involved. Chopping, therefore, needs to be watched carefully when integrating differential equations.

23.5 THE THREE-POINT PREDICTOR

The most desirable predictor to go with the corrector (Sec. 23.2) would be one that used the information at the last three points, namely,

$$y_{n+1} = A_0 y_n + A_1 y_{n-1} + A_2 y_{n-2} + h(B_0 y'_n + B_1 y'_{n-1} + B_2 y'_{n-2}) + \bar{E}_5 \frac{h^5 y^{(5)}}{5!}$$

The coefficients may be found by the usual process or by setting $b_{-1} = 0$. The results are

$$\begin{aligned} A_0 &= -8 - A_2 & B_0 &= \frac{17 + A_2}{3} \\ A_1 &= 9 & B_1 &= \frac{14 + 4A_2}{3} \\ A_2 &= A_2 & B_2 &= \frac{-1 + A_2}{3} \\ & & \bar{E}_5 &= \frac{40 - 4A_2}{3} \end{aligned}$$

The large value $A_1 = 9$ means that at best we shall have a large noise amplification

$$N_a = (A_0{}^2 + A_1{}^2 + A_2{}^2)^{1/2} > 9$$

in the predictor.

In order to keep the three-point predictor and avoid the high noise amplification, we could make the formula exact through only x^3, leaving the error term of order h^4. This leads to the family of predictors

$$\begin{aligned}
A_0 &= 1 - A_1 - A_2 & B_0 &= \frac{23 + 5A_1 + 4A_2}{12} \\
A_1 &= A_1 & B_1 &= \frac{-16 + 8A_1 + 16A_2}{12} \\
A_2 &= A_2 & B_2 &= \frac{5 - A_1 + 4A_2}{12} \\
& & \bar{E}_4 &= 9 - A_1
\end{aligned}$$

If we now use the difference of the nth predictor minus the corrector to estimate the error of the $(n + 1)$st predictor, we would modify p_{n+1} by

$$m_{n+1} = p_{n+1} - (p_n - c_n) = c_n + (p_{n+1} - p_n)$$

While such a scheme is occasionally justified, it may also be a waste of computation, especially when computing accurate solutions of differential equations, and we shall not discuss it further, although it clearly has a fifth-order error term.

23.6 MILNE-TYPE PREDICTORS

In order to have seven parameters to use in searching for a suitable predictor, we can add either one more old value of the function, namely y_{n-3}, or one old value of the derivative, namely y'_{n-3}. The first proposal includes Milne's predictor, and we call these "Milne-type predictors," while the second contains the Adams-Bashforth predictor and is named accordingly.

We therefore try first to find a formula of the form

$$\begin{aligned}
y_{n+1} = A_0 y_n + A_1 y_{n-1} + A_2 y_{n-2} + A_3 y_{n-3} + h(B_0 y'_n \\
+ B_1 y'_{n-1} + B_2 y'_{n-2}) + \bar{E}\frac{h^5 y^{(5)}}{5!}
\end{aligned}$$

The usual method gives

$$\begin{aligned}
A_0 &= -8 - A_2 + 8A_3 & B_0 &= \frac{17 + A_2 - 9A_3}{3} \\
A_1 &= 9 - 9A_3 & B_1 &= \frac{14 + 4A_2 - 18A_3}{3} \\
A_2 &= A_2 & B_2 &= \frac{-1 + A_2 + 9A_3}{3} \\
A_3 &= A_3 & \bar{E}_5 &= \frac{40 - 4A_2 + 72A_3}{3}
\end{aligned}$$

For this formula to have the indicated error term, it is necessary that the *influence function*

$$G(s) = (h - s)_+^4 - A_0(-s)_+^4 - A_1(-h - s)_+^4 - A_2(-2h - s)_+^4$$
$$- A_3(-3h - s)_+^4 - 4h[B_0(-s)_+^3 + B_1(-h - s)_+^3 + B_2(-2h - s)_+^3]$$

be of constant sign for $h \geq s \geq -3h$. The line in the A_2, A_3 plane along which $G(s) = 0$ moves up from below to the axis $A_3 = 0$ as s goes from h to $-2h$, and it remains at $A_3 = 0$ for $-2h \geq s \geq -3h$. Thus we are restricted to $A_3 \geq 0$ if the error term is to be correct.

If we try to reduce the truncation error by choosing $A_3 = 0$, then we find $A_1 = 9$, which is uncomfortably large. On the other hand, if we try to minimize the noise amplification

$$N_a = (A_0{}^2 + A_1{}^2 + A_2{}^2 + A_3{}^2)^{1/2}$$

we get

$$A_0 = -\tfrac{4}{114} \qquad A_1 = \tfrac{9}{114} \qquad A_2 = -\tfrac{4}{114} \qquad A_3 = \tfrac{113}{114}$$

This is very close to Milne's predictor

$$y_{n+1} = y_{n-3} + \frac{4h}{3}(2y_n' - y_{n-1}' + 2y_{n-2}') + \frac{14}{45}h^5 y^{(5)}(\theta)$$

with $A_0 = A_1 = A_2 = 0$ and $A_3 = 1$. The gain does not seem worth it.

Thus if we try this approach, we are led to Milne's predictor. As discussed in Sec. 22.4, this may be combined in the pattern of Chap. 22 with any suitable corrector. Of course other choices of the parameters A_2, A_3 are possible.

PROBLEMS

23.6.1 Examine predictors of the form

$$y_{n+1} = A_0 y_n + A_1 y_{n-1} + A_2 y_{n-2} + h(B_0 y_n' + B_1 y_{n-1}') + \bar{E}_4 \frac{h^4 y^{(4)}(\theta)}{4}$$

using A_2 as the parameter.

Ans. $A_0 = -4 - 5A_2 \quad A_1 = 5 + 4A_2 \quad A_2 = A_2$
$B_0 = 4 + 2A_2 \quad B_1 = 2 + 4A_2 \quad \bar{E}_4 = 4 - 4A_2$

If $A_2 = 1$, then $\bar{E}_4 = 0$ and $\bar{E}_3 = 12$.

23.7 ADAMS-BASHFORTH-TYPE PREDICTORS

If, instead of using an extra value of the function, we use an extra value of the derivative, y'_{n+3}, then we have the form

$$y_{n+1} = A_0 y_n + A_1 y_{n-1} + A_2 y_{n-2} + h(B_0 y'_n + B_1 y'_{n-1} + B_2 y'_{n-2} + B_3 y'_{n-3}) + \bar{E}_5 \frac{h^5 y^{(5)}}{5!}$$

and the values

$$\begin{aligned} A_0 &= 1 - A_1 - A_2 & B_1 &= \frac{-59 + 19A_1 + 32A_2}{24} \\ A_1 &= A_1 & B_2 &= \frac{37 - 5A_1 + 8A_2}{24} \\ A_2 &= A_2 & B_3 &= \frac{-9 + A_1}{24} \\ B_0 &= \frac{55 + 9A_1 + 8A_2}{24} & \bar{E}_5 &= \frac{251 - 19A_1 - 8A_2}{6} \end{aligned}$$

In this case the influence function $G(s)$ has zeros in the A_1, A_2 plane along a line which starts as $A_1 = 9$ for $s = -2h$ and rises while tilting through negative slopes as s increases, producing no serious restraints on the coefficients.

A number of cases are given in Table 23.7, including the Adams-Bashforth predictor. The noise propagation value N_c for $h \to 0$ is the same as for the corrector and need not be repeated here.

Table 23.7 ADAMS-BASHFORTH-TYPE PREDICTORS [THE VALUES OF $\bar{E}_5 - E_5$ ARE COMPUTED FOR THE CASE $a_i = A_i$ $(i = 0, 1, 2)$]

	Adams-Bashforth	High accuracy	Three-eighths rule	"$\frac{1}{3}$"	"$\frac{1}{2}$"	"$\frac{2}{3}$"
A_0	1	0	0	$\frac{1}{3}$	$\frac{1}{2}$	0
A_1	0	1	0	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$
A_2	0	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$
B_0	$\frac{35}{24}$	$\frac{8}{3}$	$\frac{21}{8}$	$\frac{91}{36}$	$\frac{119}{48}$	$\frac{191}{72}$
B_1	$-\frac{39}{24}$	$-\frac{5}{3}$	$-\frac{9}{8}$	$-\frac{63}{36}$	$-\frac{99}{48}$	$-\frac{107}{72}$
B_2	$\frac{37}{24}$	$\frac{4}{3}$	$\frac{15}{8}$	$\frac{57}{36}$	$\frac{69}{48}$	$\frac{109}{72}$
B_3	$-\frac{9}{24}$	$-\frac{1}{3}$	$-\frac{3}{8}$	$-\frac{13}{36}$	$-\frac{17}{48}$	$-\frac{25}{72}$
$\bar{E}_5$	$\frac{251}{6}$	$\frac{116}{3}$	$\frac{243}{6}$	$\frac{121}{3}$	$\frac{161}{4}$	$\frac{707}{18}$
$\bar{E}_5 - E_5$	$\frac{270}{6}$	$\frac{240}{6}$	$\frac{270}{6}$	$\frac{260}{6}$	$\frac{255}{6}$	$\frac{250}{6}$

PROBLEMS

23.7.1 Examine predictors of the form

$$y_{n+1} = A_0 y_n + A_1 y_{n-1} + h(B_0 y'_n + B_1 y'_{n-1} + B_2 y'_{n-2}) + \bar{E}_4 \frac{h^4 y^{(4)}(\theta)}{4}$$

using A_1 as a parameter.

$$Ans. \quad A_0 = 1 - A_1 \qquad B_0 = \frac{23 + 5A_1}{12} \qquad B_2 = \frac{5 - A_1}{12}$$

$$A_1 = A_1 \qquad B_1 = \frac{-16 + 8A_1}{12} \qquad \bar{E}_4 = 9 - A_1$$

23.8 GENERAL REMARKS ON THE CHOICE OF A METHOD

In designing a specific method, the choice of the predictor and corrector involves balancing three somewhat incompatible objectives:

1 Low truncation error
2 Large margin of stability
3 Protection against roundoff troubles

It is best to begin by selecting a corrector, since it plays the more important role. Figure 21.5.1 shows the domain from which we may select our corrector. It is clear that Milne's corrector ($a_1 = 1$, $a_2 = 0$) is about as accurate as can be found. Figure 23.3.1 shows the stability measure. If $A = \partial f/\partial y$ is positive in the entire interval of integration, then Milne is a good choice. But if $A = \partial f/\partial y$ can be negative, then Milne's corrector is unstable, and we need to decide on how we shall compromise between the two troubles, instability and truncation, which are measured, respectively, by

$$\frac{\partial f}{\partial y} h \equiv Ah \qquad \text{and} \qquad \frac{E_5 h^5 y^{(5)}(\theta)}{5!}$$

No specific relationship between these two terms exists, although there is clearly a tendency for both to be large or both to be small.

We shall arbitrarily select $Ah = -\frac{4}{10}$ as a safe amount of protection against instability troubles for moderately accurate solutions for most equations. With this choice, a comparison of Figs. 21.5.1 and 23.3.1 shows that the truncation error is about the same all along the straight side of the contour $Ah = -\frac{4}{10}$. Roundoff troubles suggest moving to the upper right and selecting the point ($a_1 = \frac{2}{3}$,

$a_2 = \frac{1}{3}$) as the appropriate corrector. In a sense, this choice is a mixture of two-thirds of Milne with one-third of the three-eighths-rule. Milne and Reynolds[1] have suggested the occasional use of the three-eighths-rule while using Milne's method. The above proposed two-thirds method is a particular such mixture (at every step as it were) and provides stability in a more satisfactory manner.

While we have selected $Ah = -\frac{4}{10}$ as a generally safe amount of protection against instability, it should be noted that for computing low-accuracy solutions of equations with large negative $\partial f/\partial y$, a more stable method should be chosen. From Fig. 23.3.1 it appears that $a_1 = 0$, $a_2 = \frac{1}{16}$ is a very stable method, but it has obviously a fairly large ($E_5 = -\frac{23}{12}$) truncation error.

PROBLEMS

23.8.1 From Prob. 23.3.2, discuss the choice of a fourth-order corrector.

23.9 CHOICE OF PREDICTOR

In choosing a predictor there is a strong temptation to use the same values for the A's in the predictor as for the a's in the corrector. For one reason, in computing trajectories to the moon or the planets, for example, we could compute the sum of the y values (positions) by a double-precision routine and probably manage to compute the y' terms, which depend on differences of position, with single precision and at the same time achieve a large increase in accuracy. This matching of the a's and A's eliminates the Milne-type predictors, which have slightly more accuracy than the Adams-Bashforth type. We shall, however, examine only the cases where they match, as this provides quite a range of systems. The reader is invited to design his own method to fit his specific case.

The difference between the predicted and corrected values of y_{n+1} measures the fifth derivative. With matching a's and A's, the y terms cancel, and we have

$$\begin{aligned} p_{n+1} - c_{n+1} = h[&B_0 y'_n + B_1 y'_{n-1} + B_2 y'_{n-2} + B_3 y'_{n-3} \\ &- (b_{-1} p'_{n+1} + b_0 y'_n + b_1 y'_{n-1} + b_2 y'_{n-2})] \end{aligned}$$

which can generally be computed quite accurately.

PROBLEMS

23.9.1 Discuss the corresponding fourth-order theory.

[1] W. E. Milne and R. R. Reynolds, Stability of Numerical Solution of Differential Equations, *J. Assoc. Computing Machinery*, vol. 6, pp. 196–203, April, 1959.

23.10 SELECTED FORMULAS

It is a common observation that $p_n - c_n$ tends to be somewhat constant from step to step so that, knowing that p_{n+1} is in error by

$$\frac{\bar{E}_5 h^5 y^{(5)}}{5!}$$

we naturally use

$$-\frac{\bar{E}_5}{\bar{E}_5 - E_5}(p_n - c_n) = -\frac{\bar{E}_5 h^5 y^{(5)}}{5!}$$

to mop up this error. (In the first predictor step, when we lack the $p_n - c_n$, we use $p_n - c_n = 0$.) A similar argument applies to the corrector where it is natural to use

$$\frac{E_5}{\bar{E}_5 - E_5}(p_{n+1} - c_{n+1}) \approx \frac{E_5 h^5 y^{(5)}}{5!}$$

as the correction to the corrector value.

Applying this to the two-thirds case as an example, we use the following procedure:

Predict:

$$p_{n+1} = \frac{2y_{n-1} + y_{n-2}}{3} + \frac{h}{72}(191y'_n - 107y'_{n-1} + 109y'_{n-2} - 25y'_{n-3}) + \frac{707}{2{,}160}h^5 y^{(5)}$$

Modify:

$$m_{n+1} = p_{n+1} - \tfrac{707}{750}(p_n - c_n)$$

Correct:

$$c_{n+1} = \frac{2y_{n-1} + y_{n-2}}{3} + \frac{h}{72}(25m'_{n+1} + 91y'_n + 43y'_{n-1} + 9y'_{n-2}) - \frac{43}{2{,}160}h^5 y^{(5)}$$

Final value:

$$y_{n+1} = c_{n+1} + \tfrac{43}{750}(p_{n+1} - c_{n+1})$$

In this process we get an exact answer if $y^{(5)}(x) =$ constant, so that it may be called a sixth-order method.

It should be noted that if y_{n+1} is close to m_{n+1}, then the second evaluation of the derivative to get y'_{n+1} *need not* be done. If ε is the difference, then it will result in (assuming $|Ah| = 4/10$)

$$\varepsilon \cdot h \frac{191}{72} \left| \frac{\partial f}{\partial y} \right| \leq \varepsilon \cdot 2.65 |A| h \approx 1.06\varepsilon$$

in the next predicted value.

Another case worth examination is the "one-half" which is an average of the Milne and Adams-Bashforth cases. Again arbitrarily using matching coefficients,

$$p_{n+1} = \frac{y_n + y_{n-1}}{2} + \frac{h}{48}(119y'_n - 99y'_{n-1} + 69y'_{n-2} - 17y'_{n-3}) + \frac{161}{480} h^5 y^{(5)}$$

$$m_{n+1} = p_{n+1} - \tfrac{161}{170}(p_n - c_n)$$

$$c_{n+1} = \frac{y_n + y_{n-1}}{2} + \frac{h}{48}(17m'_{n+1} + 51y'_n + 3y'_{n-1} + y'_{n-2}) - \frac{9}{480} h^5 y^{(5)}$$

$$y_{n+1} = c_{n+1} + \tfrac{9}{170}(p_{n+1} - c_{n+1})$$

This has a slightly smaller truncation error than the "two-thirds" method. However, $N_c = 2/3$ and $N_a = 0.7071$, versus $N_c = 3/5$ and $N_a = 0.75$ for the two-thirds method.

PROBLEMS

23.10.1 Discuss the corresponding fourth-order theory and develop specific formulas.

23.11 DESIGNING A SYSTEM[1]

Besides the formulas for integrating each step, it is necessary to have formulas for halving and doubling the interval and criteria for when to do so. We also need a starting routine which we shall here assume is the Runge-Kutta method (see Chap. 24).

There are two possible reasons for halving (or subdividing in any other way) the step size of the integration: One is a large truncation error; the other is instability. Protection against instability during a computation depends on the

[1] For a good discussion see Arnold Nordsieck, On Numerical Integration of Ordinary Differential Equations, *Math. of Comp.*, vol. 16, no. 77, pp. 22–49, January, 1962, and Gear [16].

ability to estimate $A = \partial f/\partial y$. If we evaluate the derivatives for both m_{n+1} and y_{n+1}, we get

$$f(x_{n+1}, m_{n+1}) - f(x_{n+1}, y_{n+1}) \approx \frac{\partial f(x_{n+1}, y_{n+1})}{\partial y}(m_{n+1} - y_{n+1}) \approx A(m_{n+1} - y_{n+1})$$

Note that an accurate estimate of A is not possible from this formula. The occasional estimation of A by doing the two evaluations in a single step is necessary in order to be safe from instability troubles if some other method of estimating A is not available.

If A is found to be too large for the current step size h, then we can either decrease h or change to a more stable formula. If at the same time the truncation error is low, then the change of formula to a more stable one with a larger truncation error is probably better. On the other hand, if both are close to tolerance, then h should be decreased.

We have repeatedly indicated that the truncation error is measured by

$$p_n - c_n = \frac{(\bar{E}_5 - E_5)h^5 y^{(5)}}{5!}$$

Actually, this is not so in the proposed systems, since, as noted, if $y^{(5)}$ were constant, then the truncation error would be zero. The true error is measured by the change in $p_n - c_n$ from step to step. The first temptation is to compute

$$(p_{n+1} - c_{n+1}) - (p_n - c_n)$$

and use it as a guide for when to halve the interval. Unfortunately, as a few moments' reflection will reveal, such a quantity is extremely dependent upon the local "noise" in the computation (except in the case of low-accuracy computations done on a high-accuracy computer) and does not provide a satisfactory criterion for when to halve the interval size. We are, therefore, left in the unfortunate position of knowing that the truncation error per step is a good deal less than the quantity $p_n - c_n$, but we do not know how much.

To find a suitable halving formula, we have y_n, y_{n-1}, y_{n-2} and their derivatives, so that we can estimate $y_{n-1/2}$ from them while making the formula exact for $1, x, \ldots, x^5$ (which makes the error depend on $y^{(6)}$). Such a formula is

$$y_{n-1/2} = \tfrac{1}{128}[45y_n + 72y_{n-1} + 11y_{n-2} + h(-9y'_n + 36y'_{n-1} + 3y'_{n-2})]$$

To obtain $y_{n-3/2}$, we can reverse the formula to get

$$y_{n-3/2} = \tfrac{1}{128}[11y_n + 72_{n-1} + 45y_{n-2} - h(3y'_n + 36y'_{n-1} - 9y'_{n-2})]$$

When it comes to doubling, we can do one of the following:

1 Carry extra back values

2 Restart

3 Use special formulas for two steps